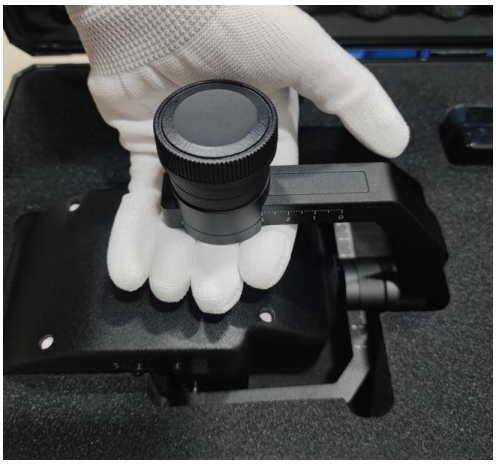
机载高光谱系列

用户使用指南

1. 收到仪器请先根据下列说明进行初始化设置

1.1. 将相机安装到无人机上

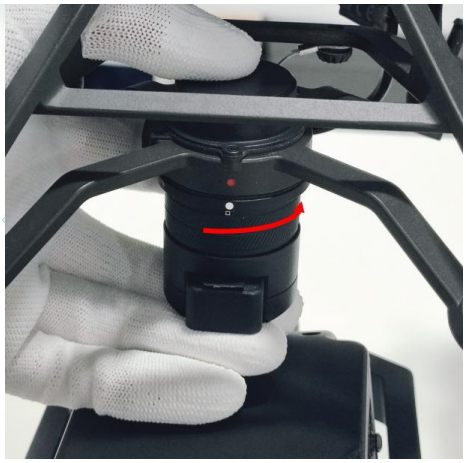
抓起云台顶部，手扶云台顶部，取出相机，如图

取下镜头保护盖



逆时针旋转云台保护盖并取下，将白点朝自己，扣到无人机云台支架上，旋转云台使云台红点对准无人机云台支架的红点

相机开机

1、打开无人机遥控器、打开无人机

2、等待相机自传完成并回到中间位置，在3s内按下相机开关

3、随后等待相机开机，相机开机后，遥控器右下角会出现名为“Figspec”的小窗口，



相机开机，软件会进行自启动，软件启动后，在右侧状态栏，相机和无人机 都显示已连接，说明相机已经正确开机

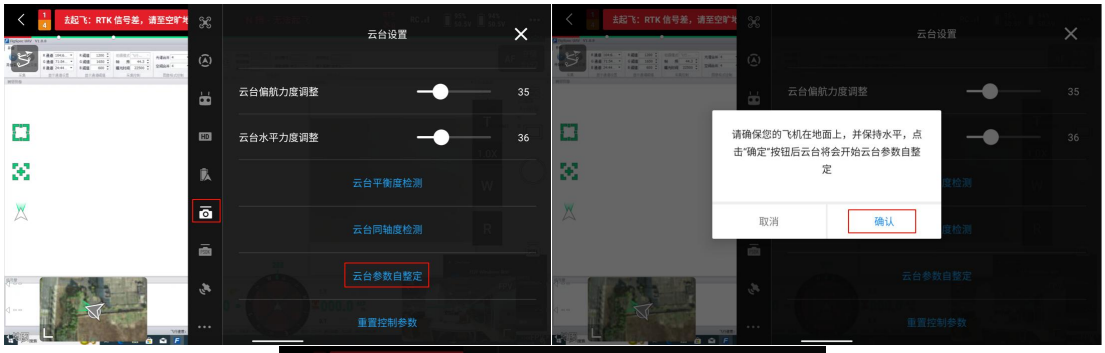


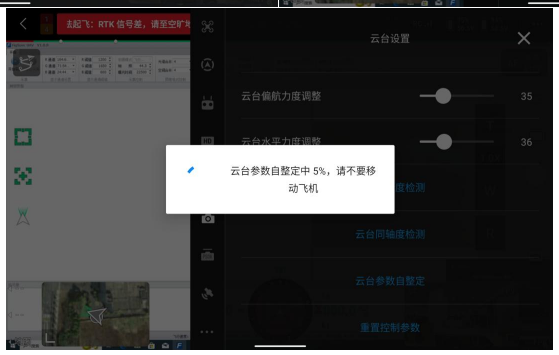
1.2. 云台参数自整定

点击界面右上角三个点，进入设置界面



电机云台图标，向上滑动屏幕至最底，点击“云台参数自整定”，云台会自行整定



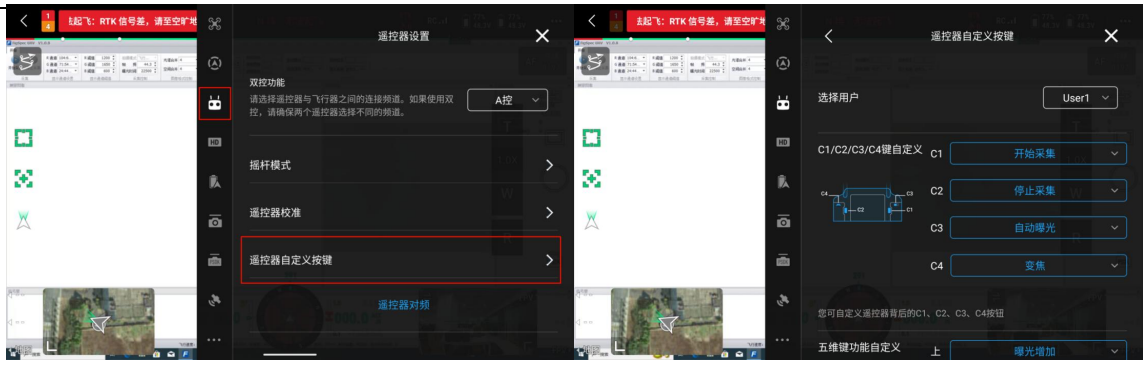


等待自整定完成即可

1.3. 快捷键设置

注：快捷键设置需要在连接高光谱相机后进行

1、点击遥控器界面右上角三个点，进入设置界面，依次选择“遥控器图标”-“遥控器自定义按键”，进入到快捷键设置界面



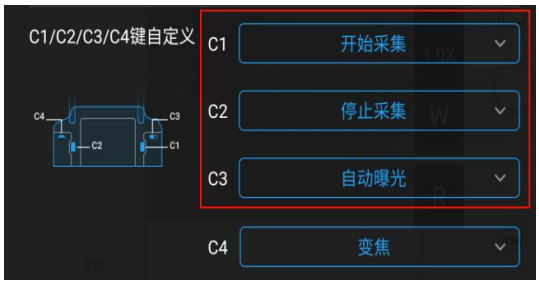
2、在图中，将对应的按钮设置为适合自己的快捷键

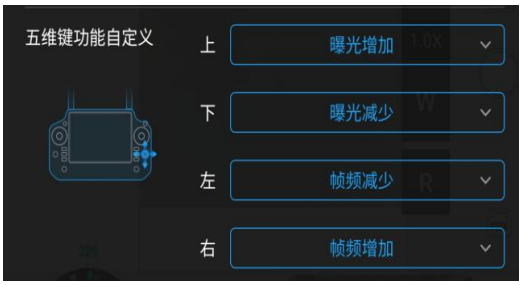
点开下拉框，左右滑动顶部选择功能区，找到PSDK，然后选择对应的功能即可



依次设置所有按键快捷方式

3、快捷键设置模版





4、M300与M350 按键图示

M300 遥控器按键图示





M350 遥控器按键图示





2. 产品参数

2.1. 机载高光谱 相机参数

照明方式	被动照明（不含光源）
分光方式	透射光栅
光谱范围	400-1000nm
光谱波段	300、400、1200
光谱分辨率(FWHM)	2.5nm
狭缝宽度	25um
透射效率	＞60%
杂散光	＜0.5%
空间像素数	480、960、1920（软件可设置）
像素大小	5.86um
成像速度	全波段128Hz,ROI后可实现3300Hz
探测器	CMOS
SNR(Peak)	600/1
相机输出	USB3.0或千兆网
相机接口	C-Mount
配件	USB3.0传输线或千兆网传输线
ROI	多个区域
嵌入式数据采集处理存储单元	N5105处理器 1T/512G SSD存储



- 操作方便，无需专业无人机操控手，可实现单人操作
- 通过地面站实时观测飞机采样地点并可利用地面站设置逐点采集的航线数据预览及矫正功能：辐射度校正、反射率校正、区域校正支持批处理
- 实时常用植被指数计算功能
- 支持自定义实时分析模型输入功能
- ENVI多种数据格式完美兼容

2.2. 机载多光谱系列 相机参数

光谱通道数量	30	60	90	120	150	180
光谱通道波长	400-1000nm每20nm输出一个波长	400-1000nm每10nm输出一个波长	400-1000nm每6.66nm输出一个波长	400-1000nm每5nm输出一个波长	400-1000nm每4nm输出一个波长	400-1000nm每3.33nm输出一个波长
光谱分辨率/半波宽	3.5nm	3.5nm	3.5nm	2.5nm	2.5nm	2.5nm
空间分辨率	1920					
采样速度	128 line/S					
图像传感器	1/1.1英寸CMOS					
有效像素	1920					
快门类型	全局快门					
量化位数	12bit					
视场	25.36°					
地面分辨率	2.8cm@h120m					
覆盖宽度	54m@h120m					
光学窗口	高透过率光学玻璃窗口					
主机尺寸	≤155*95*119mm					
主机重量	≤840g					
安装/供电接口	X-Port					
功耗	45w					
图片格式	12bit的.spe(兼容envi等第三方分析软件）					
数据存储空间	512SSD					
应用软件	FIGSPEC UAV,FIGSPEC Merage拼图软件,FIGSPEC Studion图像分析软件					
拍摄方式	实时采集					

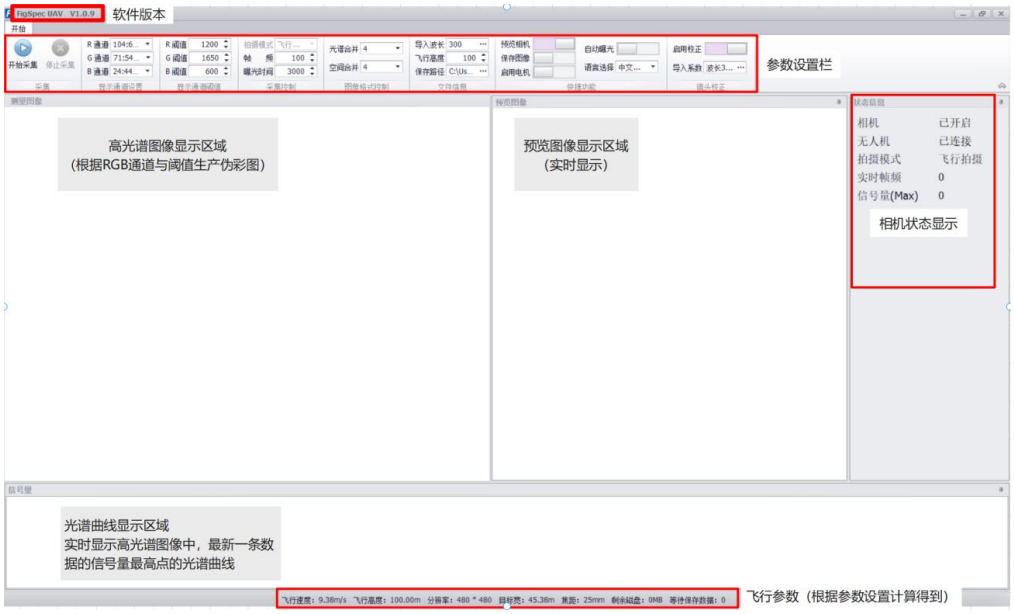


- 超高光谱通道数：30-180个光谱通道(不同型号)
- 2K空间分辨率
- 全局快门，12bit高精度采样数据
- 地面站实时预览采集数据
- DJI X-Port控制和供电、512GSSD大容量存储
- 大疆M350/M300 RTK无人机定制，即插即用
- FIGSPEC UAV实时飞控软件，FIGSPEC Merage拼图软件，FIGSPEC Studion图像分析软件

3. 软件介绍

3.1. 采集软件FigSpec UAV介绍

软件界面及参数介绍



FigSpec UAV 界面主要分为6部分：

1、参数设置栏

包含了相机所用到的所有参数的设置

2、高光谱图像显示区域

根据参数中RGB通道和RGB阈值的选择，生成高光谱伪彩图

3、预览图像显示区域

实时显示预览相机的画面，未真彩图

4、相机状态栏

显示相机的连接状态，采集方式，采集状态信息

5、光谱曲线显示区域

实时显示最新一条数据上信号量最高点的光谱曲线

6、飞行参数栏

显示飞行参数，这里的参数是实际执行任务时应用的参数（根据参数设置栏计算得到）

3.2. 参数设置栏介绍



1、开始采集

控制相机开始采集高光谱图像（设置快捷键为C1）

2、结束采集

控制相机停止采集（设置快捷键为C2）



3、RGB通道设置

① 波长范围为400-1000nm的相机，建议设置为：

R:104-610nm R 通道 104:6... ；

G:71-542nm G 通道 71:54... ；

B:24-443nm B 通道 24:44... ；

这三个通道合成的高光谱伪彩图更接近真彩色

② 波长范围为900-1700nm的相机，建议设置为1000-1600nm之间的单通道值：

900-1700nm为肉眼不可见光，没有合适的彩色RGB通道值推荐，所以设置为灰度图

如R:500-1300nm; G:500-1300nm; B:500-1300nm。

4、RGB阈值设置

RGB阈值根据图像的信号量（即：高光谱伪彩图的亮暗）进行设置，

如果图像偏暗，则将阈值调低；

如果图像偏亮，则将阈值调高。

波长范围为400-1000nm的相机，大致比例建议设置为：

R:G:B = 5:6:4；例如：R:G:B = 800:1000:600

波长范围为900-1700nm的相机，根据实际图像设置统一即可：

R:G:B = 1:1:1；例如：R:G:B = 1000:1000:1000



5、采集参数设置

① 拍摄模式：

- 飞行模式：FS50系列、FS60C、FS62C均默认选择飞行模式，并不可修改（无人机推扫采集模式）

- 悬停模式：定制型号的机载高光谱相机，可以支持悬停拍摄（即无人机悬停，相机面扫一块区域的高光谱图像），但是这种模式不支持DJI M300和M350RTK无人机，可定制适配 DJI M600 PRO

② 帧频：

相机每秒的采集速度，单位Hz。即相机每秒采集多少帧图像

③ 曝光时间：

相机测量1次（1帧）接收光的时间，单位是us

曝光时间需要根据实际采集时光照强度不同进行设置，具体调试方式后面讲解



6、图像格式设置

① 光谱合并

光谱通道上的合并，即采集数据时每个像素点上光谱数据的波段数，参数详见下表

② 空间合并

图像空间像素数的合并，即图像高度像素数量的多少，参数详见下表

这里只列出图像高度分辨率，图像的长度是推扫出来的

	合并模式	1合并	2合并	4合并
FS60C 空间合并(1920)	求和	1920 px	960 px	480 px
FS60C 光谱合并(1200)	求和	1200	600	300
FS50 空间合并（1920）	求和	1920 px	960 px	480 px
FS50 光谱合并		不支持	不支持	不支持
FS62 空间合并 (1280)	求和	1280 px	640 px	320px
FS62 光谱合并（1024）	求和	1024	512	256



7、飞行参数设置

① 导入波长：

每台相机拥有唯一的配置文件，相机之间不可互用配置文件。

出厂进行默认配置，并在相机内进行了备份，文件格式为“.chnwave”

② 飞行高度：

无人机飞行的高度，单位：米（m）

飞行高度根据个人的实际需要进行设置

飞行高度会影响飞行速度和目标分辨率（即单个像素对应的实际空间大小）

无人机航线任务内的飞行高度需要和这里的参数保持一致

③ 保存路径：

“保存路径”是保存高光谱图像和预览图像的目录，出厂进行了默认配置，一般为 ‘C:\FigSpec\file\’ 或 ‘D:\FigSpec\file\’ 文件夹

保存的周期为：完整采集一次（开始采集到停止采集）；

注：只开始采集，不停止采集，数据不会保存。



8、其他功能设置

① 预览相机：

默认打开，实时显示相机采集的区域

② 保存图像：

默认关闭，保存的是预览相机拍摄的画面，拍照频率为 1 张/秒

③ 启用电机：

默认关闭

FS60C、FS50系列、FS62C不支持启用电机

定制机载相机（M600 PRO挂在23或27）支持启用电机（即支持悬停模式）



9、镜头校正

① 启用校正：

镜头校正是对相机本身对光照产生的影响进行校正

出厂默认启用，并导入了对应的镜头校准系数

② 导入系数：

导入镜头的校准系数，出厂进行默认配置，并在相机内进行了备份



10、其他参数设置

① 自动曝光：

默认关闭，为防止飞行过程中误触导致数据报废，建议不启用

启用后，在遥控器上点击自动曝光，相机会自动计算出合适的曝光时间并设置，不再需要人为的去调试

② 语言选择：

实现中英语言系统的切换，出厂默认根据系统语言进行设置

3.3. 状态信息介绍



1、相机：显示相机的连接状态
(1) 已开启表示可以使用
(2) 未连接表示无法使用
2、无人机：无人机的连接状态
(1) 已连接表示可以通过无人机控制相机
(2) 未连接表示无法通过无人机控制相机
3、拍摄模式：
显示当前选择的拍摄模式
4、实时帧频
显示采集时，实时的采集帧频
5、信号量(max)：
显示实时采集到的数据中信号量最大的值，单位%

3.4. 飞行参数

在参数设置栏进行参数设置后，软件自动计算出标准的飞行参数并显示在软件底栏

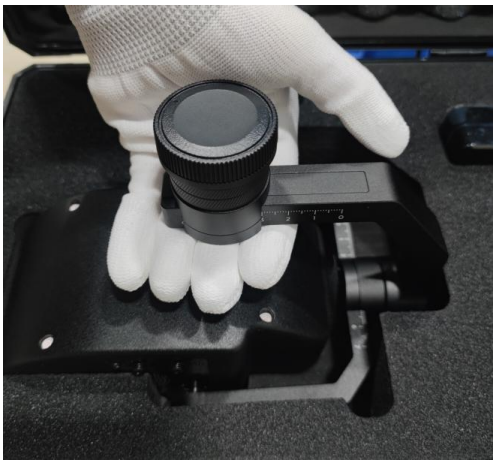


1、飞行速度 m/s：推荐的无人机飞行速度，无人机规划航线任务时，速度需要与这里的速度保持一致
2、飞行高度 m：无人机航线任务内的飞行高度需要和这里的参数保持一致
3、分辨率 px：主要指高光谱图像的高度
4、目标宽 m：相机可以采集到的实际宽度
5、焦距 mm：根据相机配置默认设置，不可更改
6、剩余磁盘 MB：显示磁盘空间，磁盘空间不足时保存会失败
7、等待保存的数据：显示没有被保存的数据量，显示0说明全部保存

4. 实飞流程操作演示

4.1. 相机安装

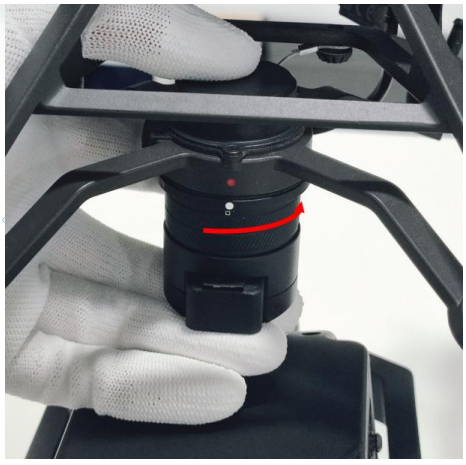
抓起云台顶部，手扶云台顶部，取出相机，如图

取下镜头保护盖



逆时针旋转云台保护盖并取下，将白点朝自己，扣到无人机云台支架上，旋转云台使云台红点对准无人机云台支架的红点

4.2. 相机与无人机开机

1、打开无人机遥控器、打开无人机

2、等待相机自传完成并回到中间位置，在3s内按下相机开关

3、随后等待相机开机，相机开机后，遥控器右下角会出现名为“Figspec”的小窗口



相机开机，软件会进行自启动，软件启动后，在右侧状态栏，相机和无人机 都显示已连接，说明相机已经正确开机



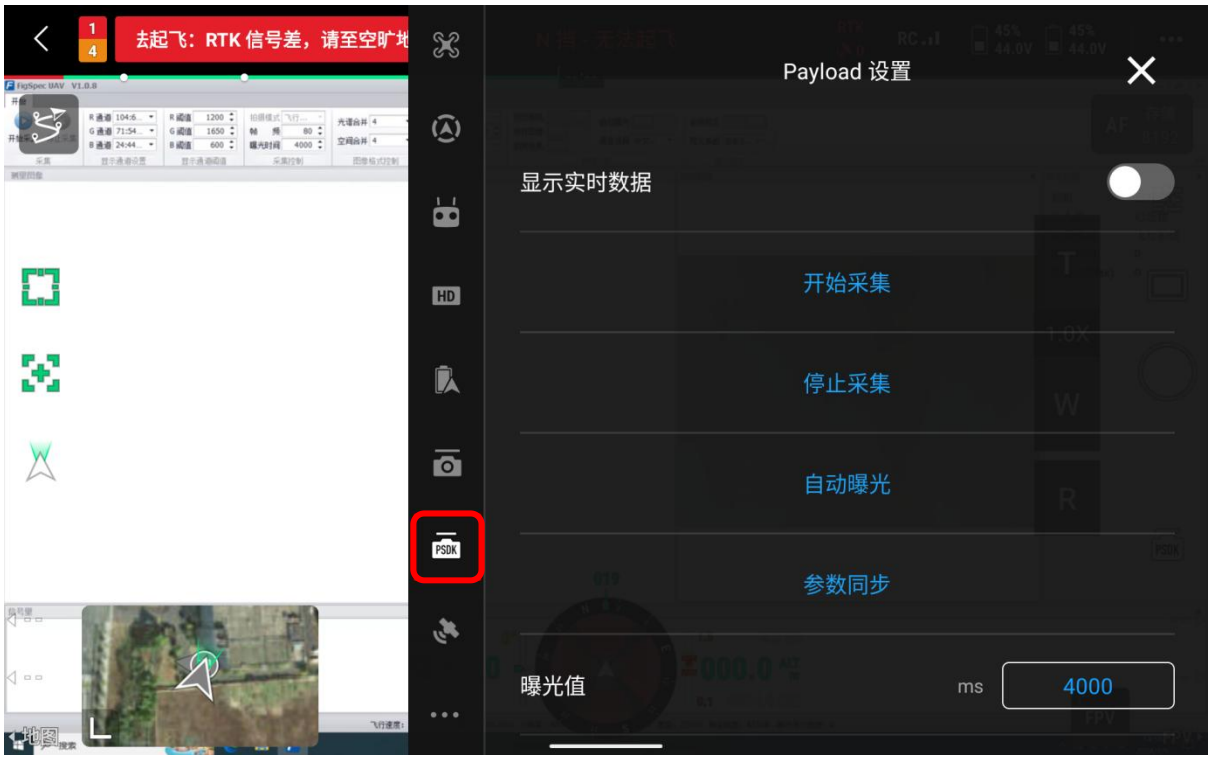
4.3. 采集参数设置

4.3.1. 预设参数如图



4.3.2. 设置方式

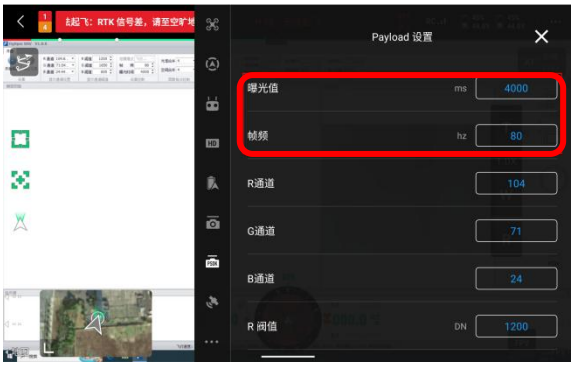
1、点开遥控器右上角三个点，找到PSDK，如图

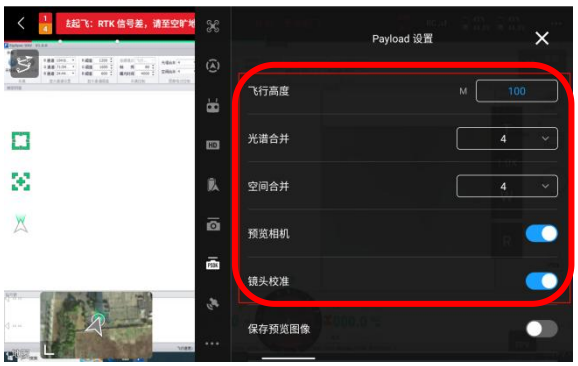


2、在PSDK中向上拖动，选择参数进行设置

主要设置 曝光值、帧频、飞行高度、光谱合并、空间合并，确认预览相机和镜头校准选项打开（这些参数在进行采集任务过程中是不可以修改的）

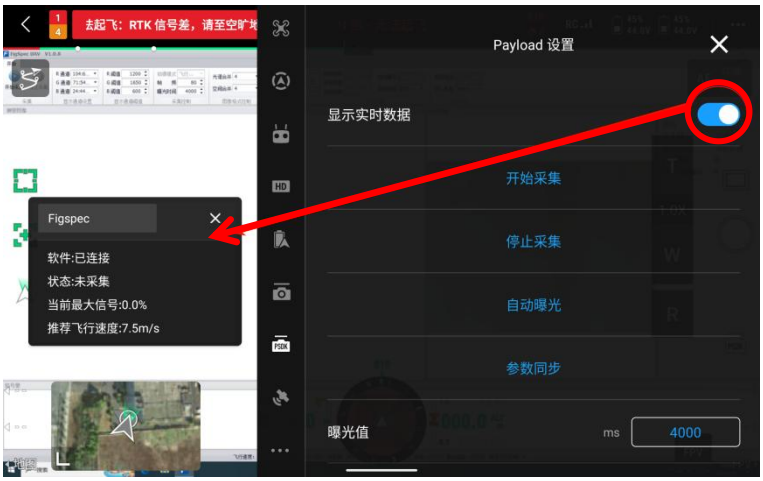
RGB阈值和通道先不设置，在采集任务过程中也可以修改





实时显示的功能

打开后弹出Figspec小窗口，显示软件连接状态、相机采集状态、最大信号（调节曝光时间使用）、推荐飞行速度（规划航线使用）



3、曝光值调节方式：

将“反射率校准布”放在相机下方，相机正下方部分不要有阴影，点击开始采集，查看“Figspec”小窗口中“当前最大信号”的值，通过调节曝光值，使最大信号量达到下表对应的值：

反射率校准布值	最大信号量值	采集区域类型
80%漫反射布	75%-85%	非水域使用
70%漫反射布	65%-75%	非水域使用
18%漫反射布	50%-70%	采集水域使用
90%漫反射板	80%-90%	非水域使用



根据软件参数设置，软件底部会计算出适合的测绘航拍参数



4.3.3. 设置步骤

1、首先，调整曝光值，使最大信号量达到白板对应的范围，如上表所述；
2、然后设置其他参数：

(1) 帧频：1-128fps

为了避免无人机飞得太快，建议将“帧速率”设置为50-100之间。

(2) 飞行高度：　　默认80

根据您自己的需求进行设置，同时确保安全。

推荐的“飞行高度”是50-150米。　　实际是80-120

(3) 空间合并&光谱合并

根据您自己的需求进行设置。

建议设置为4或2。

注意：

如果风速太高，请降低帧频或降低飞行高度。

如果风速大于5级，不建议进行采集任务。

3、现在根据以下参数进行设置：

帧速率：80fps　　飞行高度：100米

光谱合并：4　　空间合并：4

根据软件参数设置，软件底部将计算适当的航线飞行参数



4.4. 规划航线并设置参数

(1) 打开遥控器上的“地图”，点击左上角的图标  新建航线任务，选择右侧的“建图航拍”；





在地图中依次点击4个点，画出一个矩形区域，然后单击左上角的"√"；



选择无人机类型



选择相机，选择“自定义相机”来增加高光谱相机



根据图示新建相机



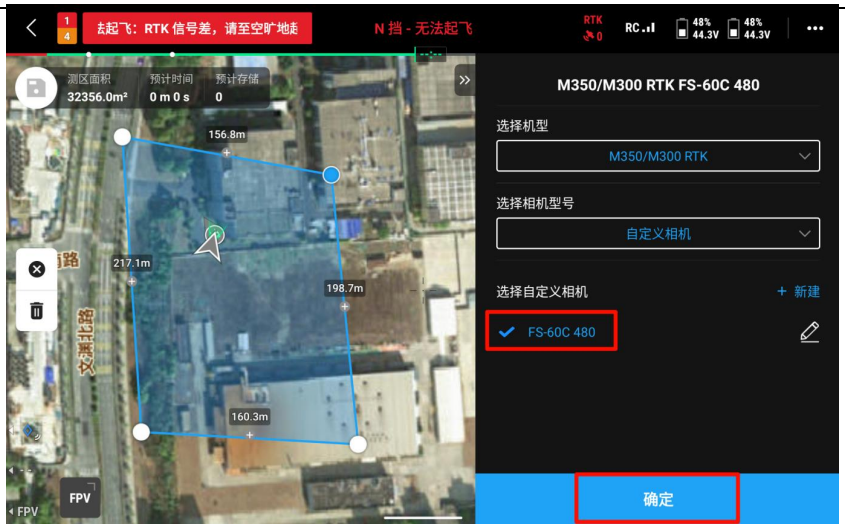
- 名称：根据相机类型和分辨率命名
- 照片大小：分辨率是根据相机软件下计算的参数填写的



- 传感器尺寸：传感器的尺寸固定为11.3mm
- 焦距：25mm

设置好后单击“完成”来保存相机。

保存好后选中添加好的相机，然后单击确定进行下一步。

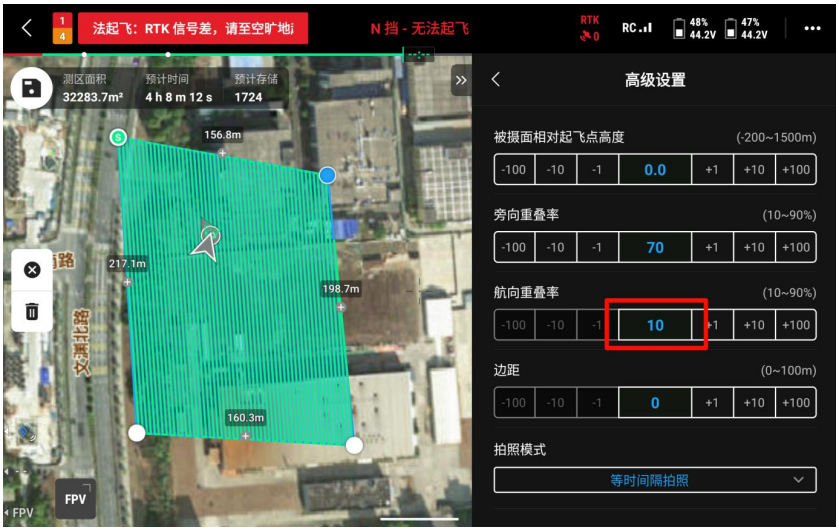


定义航线的名称



“高度模式”：默认使用相对起飞点高度

在界面最下方，进入“高级设置”，将“航向重叠率”设置为“10%”



起飞速度，根据当时天气情况而定

如果风很小或者没有风，则不需要修改此值；

如果风速较大，安全起见，将起飞速度设置在5m/s以下



设置好后返回上一级





4.5. 实飞操作

4.5.1 起飞前检查无人机状态
A、检查无人机机翼固定扣
B、检查无人机脚架固定扣
C、检查无人机电池固定扣
D、检查无人机正上方是否有障碍物
E、检查无人机周围3米内是否有障碍物
F、打开遥控器内RTK



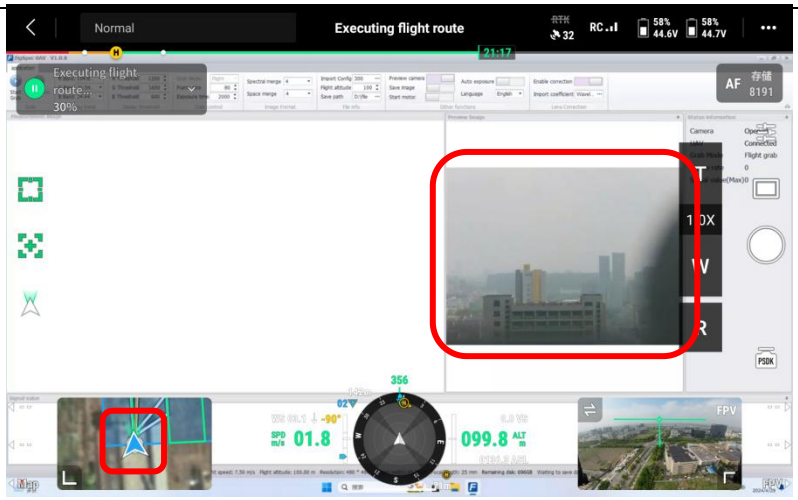



4.5.2 开始执行航线任务
点击左上角图标开始执行航线任务

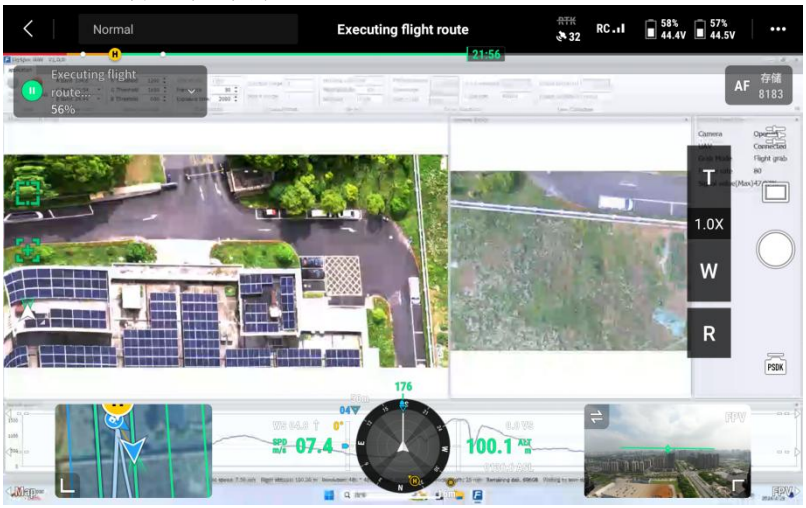
检查无人机飞行检查，如果不没有报红则表示可以正常起飞，如果提示“航向重叠率过低”，这个没关系，可以继续执行飞行任务
然后“上传航线”-“执行航线任务”
起飞时，人眼关注无人机起飞状态，时刻观察无人机飞行路线是否安全，待无人机飞远后，观察遥控器来判断无人机状态是否正常，界面切换到相机界面，观察预览相机




无人机到达航线起点后，会有语音提示“到达航线起点，开始执行任务”，相机会向上旋转90度，可以观察到相机拍摄的是空中，此时点击“开始采集”（或快捷键C1），相机会转向下方进行采集任务，在高光谱图像界面会有实时采集的高光谱图像刷新



通过观察遥控器界面查看相机采集的画面，



采集过程中如果画面太暗，则调节“PSDK”内的RGB阈值，将数值调低。如果太亮则调高。

采集数据时，需要将“反射率校准布”放置在航线范围内。



需要注意的是：在相机采集过程中，不要误触“停止采集”（或者C2）（一旦误触，则需要重新采集）

航线任务执行完成后，遥控器会有语音提示，提示后点击“停止采集”（或者C2），之后就等待无人机自动返航即可。

无人机回到人眼视线内后（100米内），双手拿好遥控器，人眼关注无人机周边，确保无人机可以安全降落；

如果遇到突发情况，单击左下角的“H”返航键，之后无人机取消自动返航并悬停在空中，突发情况解除后，可以手动将无人机操控回来，或者长按左下角“H”返航键，无人机将继续自动返航



无人机回来后，进行其他采集任务或关闭相机与无人机

关机顺序：

1、先关闭相机，单按相机的电源键，等待相机的界面完全退出后，再关闭无人机

2、关闭无人机

3、关闭遥控器

4.6. 数据拷贝

相机内置了一个电脑主机，系统为windows，可以将相机当做电脑主机使用（不建议直接处理数据）

1、给相机外接显示屏

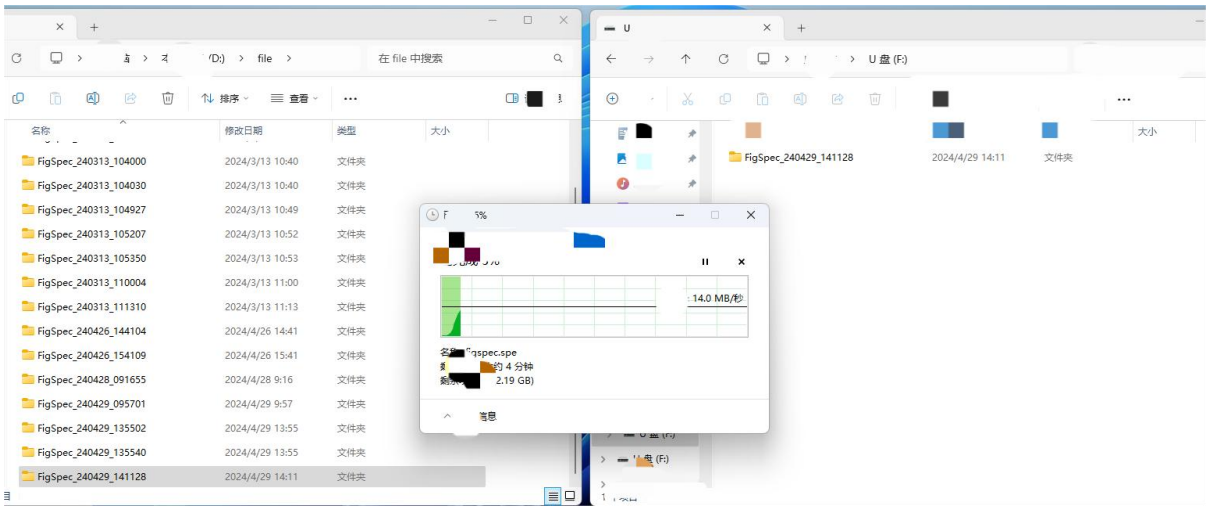




2、外接USB分线器，在分线器上连接鼠标、键盘、移动硬盘



3、打开数据保存的文件夹，将采集的数据拷贝到移动硬盘内



4、随后关闭相机，数据在自己电脑上进行数据处理

Airborne Hyperspectral Series

Manual

Catalogue

Airborne Hyperspectral Series 27

1. Initialization Settings 29

1.1. Mounting the Camera onto the Drone 29

1.2. Gimbal Parameter Auto-tuning 31

1.3. Shortcut Key Settings 32

2. Product Parameters 34

2.1. FS60C Camera Specifications 34

3. Software Introduction 35

3.1. Introduction to FigSpec UAV Acquisition Software 35

3.2. Introduction to Parameter Settings Column 36

3.3. Introduction to Status Information 39

3.4. Flight Parameters 39

4. Demonstration of Real Flight Process Operations 41

4.1. Camera Installation 41

4.2. Camera and Drone Power On 42

4.3. Data Collection Parameter Settings 43

4.4. Plan Flight Route and Set Parameters 46

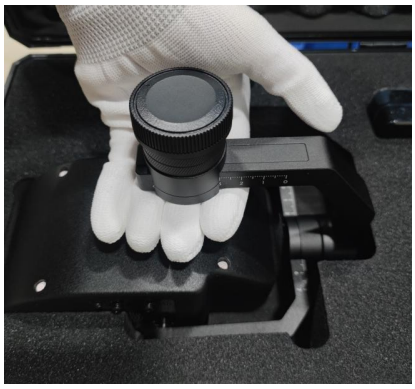
4.5. Real Flight Operations 52

4.6. Data Copying and Preprocessing 57

5. Initialization Settings

5.1. Mounting the Camera onto the Drone

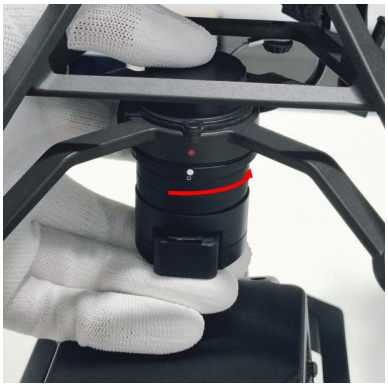
Grasp the top of the gimbal, support it with your hand, and remove the camera, as shown in the diagram.

Remove the lens cap.



Rotate the gimbal protector counterclockwise and remove it. With the white dot facing towards you, attach it to the gimbal bracket of the drone. Rotate the gimbal until the red dot on the gimbal aligns with the red dot on the drone's gimbal bracket.

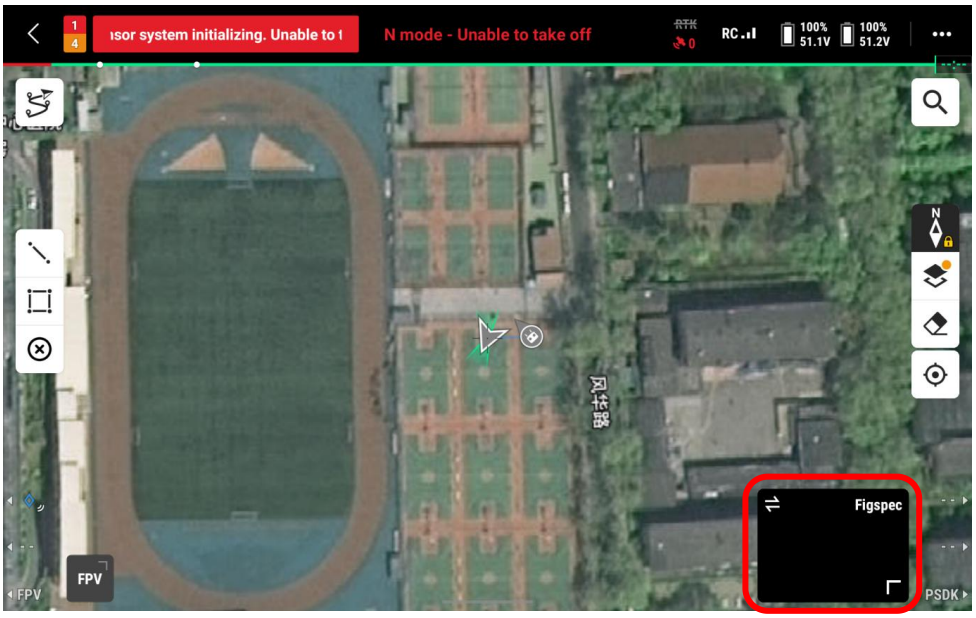



Camera Power On

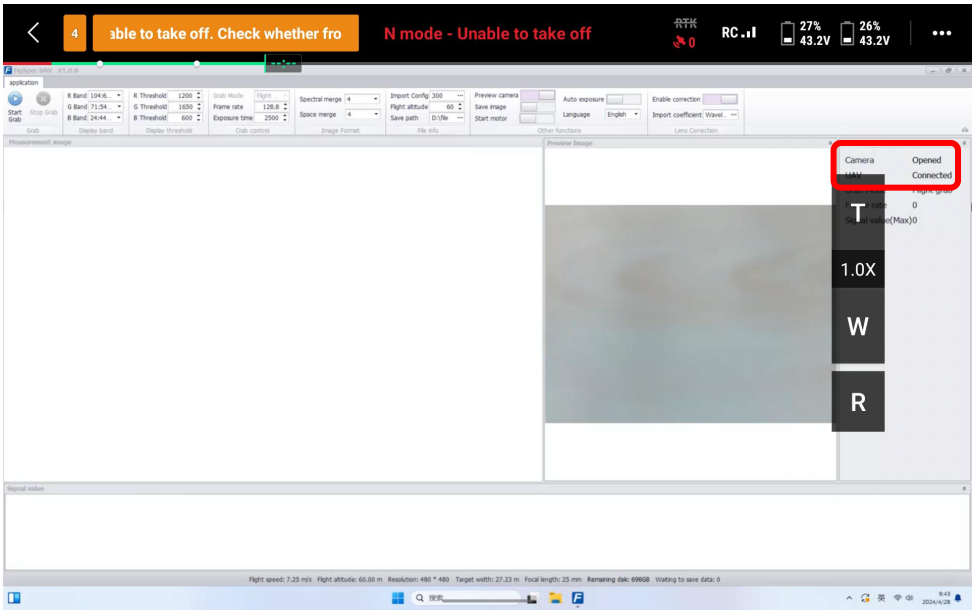
1. Turn on the drone remote controller and power on the drone.

2. Wait for the camera self-check to complete and return to the center position. Press the camera power switch within 3 seconds.

3. Wait for the camera to power on. Once the camera is powered on, a small window named Figspec will appear in the bottom right corner of the remote controller screen.

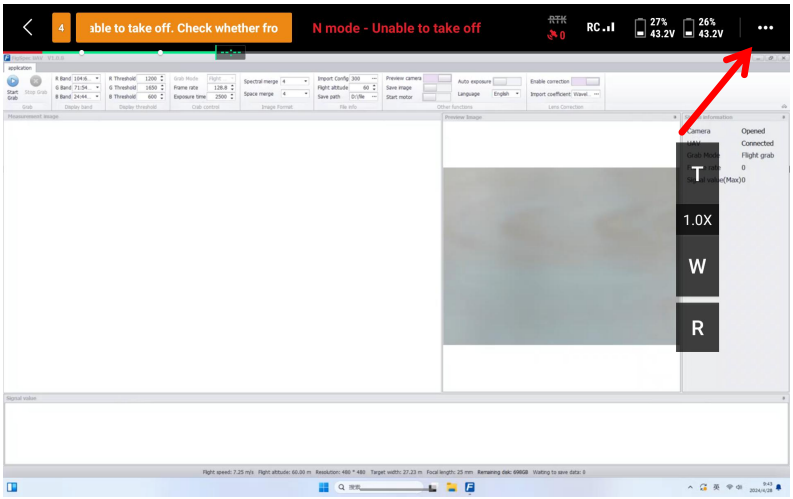


After the camera is powered on, the software will automatically start up. Once the software is launched, in the right-hand status bar, both the camera and the drone will be displayed as connected, indicating that the camera has been powered on correctly.

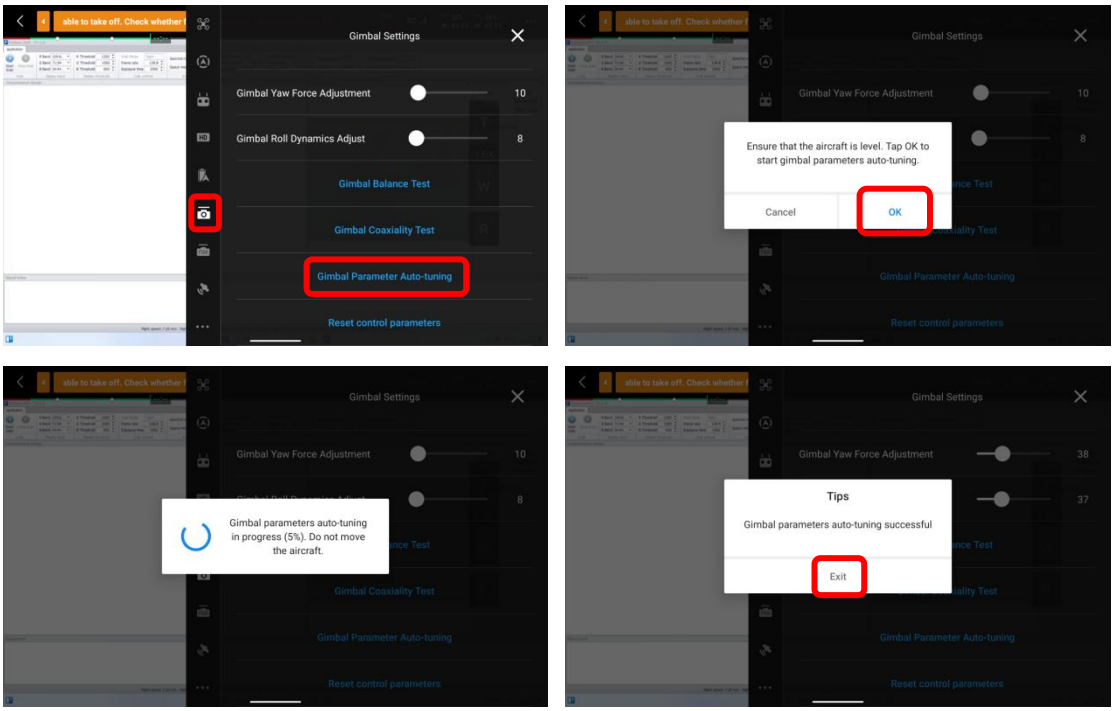


5.2. Gimbal Parameter Auto-tuning

Click on the three dots in the top right corner of the interface to access the settings menu.



Click on the gimbal icon, then swipe up on the screen until reaching the bottom, and tap on "Gimbal Parameter Auto-adjust". The gimbal will then auto-adjust itself.

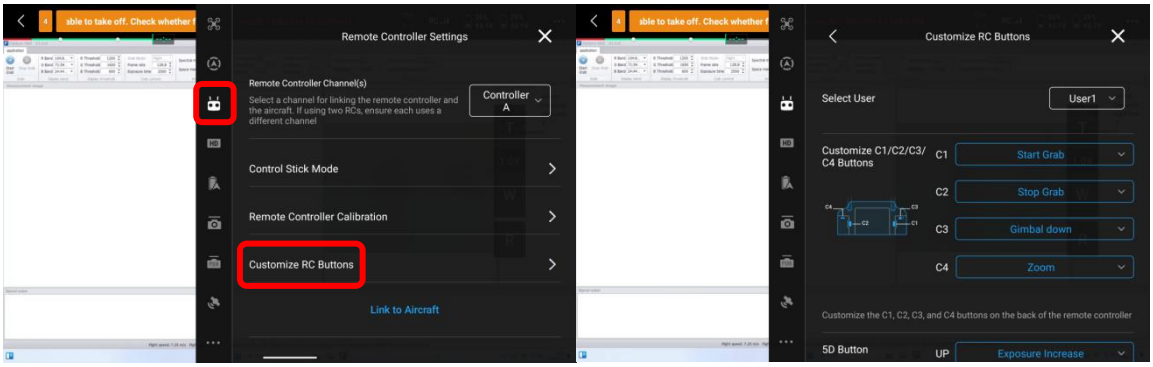


Wait for the Auto-tuning to complete.

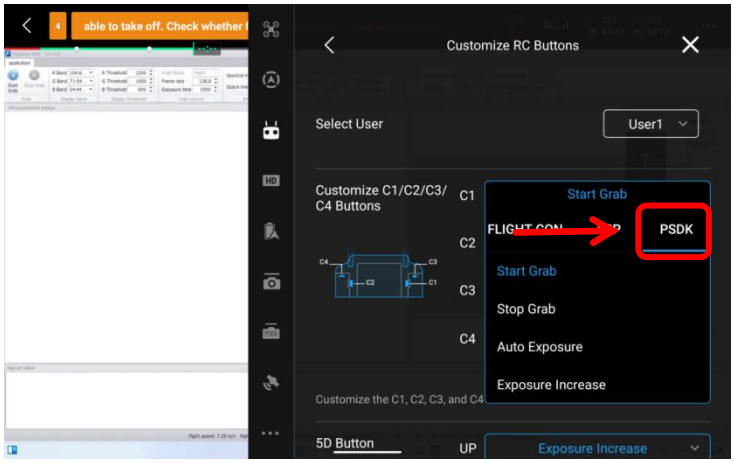
5.3. Shortcut Key Settings

Note: Shortcut key settings need to be done after connecting the hyperspectral camera.

1. Click on the three dots in the top right corner of the remote controller interface. Then, select “Remote Controller Icon” and proceed to “Customize RC Buttons” .

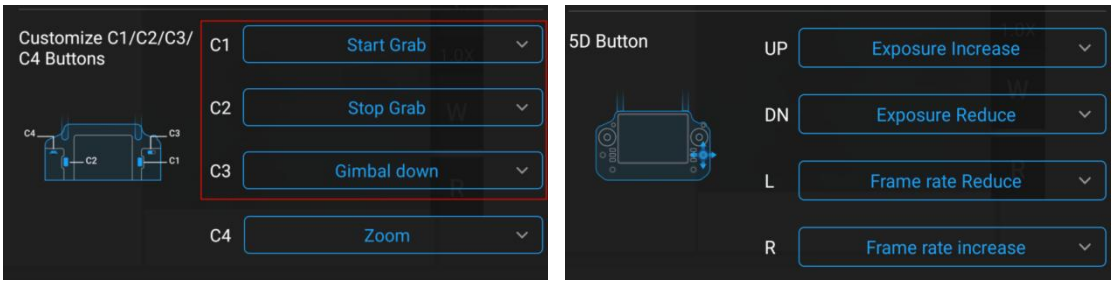


2. Set the corresponding buttons to your preferred shortcuts. Click on the dropdown menu, swipe left or right at the top to select the functional area, find “PSDK”, and then choose the corresponding function. For example, refer to “Shortcut Key”.



Configure other shortcut functions for instrument settings on different buttons.

3. Shortcut Key Settings Template.



4. M300 and M350 Button Illustrations

Remote Controller Button Illustrations for M300





Remote Controller Button Illustrations for M350





6. Product Parameters

6.1. Airborne hyperspectroscopy Camera Specifications

Lighting Mode	Passive lighting (without light source)
Spectroscopic Method	Transmission grating
Spectral Range	400-1000nm
Spectral Band	300、400、1200
Spectral Resolution (FWHM)	2.5nm
Slit Width	25um
Transmission Efficiency	＞60%
Stray Light	＜0.5%
Number of Spatial Pixels	480, 960, 1920 (software configurable)
Pixel Size	5.86um
Imaging Speed	Full band 128Hz,after ROI can achieve 3300Hz
Probe	CMOS
SNR(Peak)	600/1
Camera Output	USB3.0 or Gigabit network
Camera Interface	C-Mount
Attachment	USB3.0 transmission line or Gigabit network transmission line
ROI	Multiple regions
Embedded Data Acquisition Processing Storage Unit	N5105 processor 1T/512G SSD storage

Region of Interest

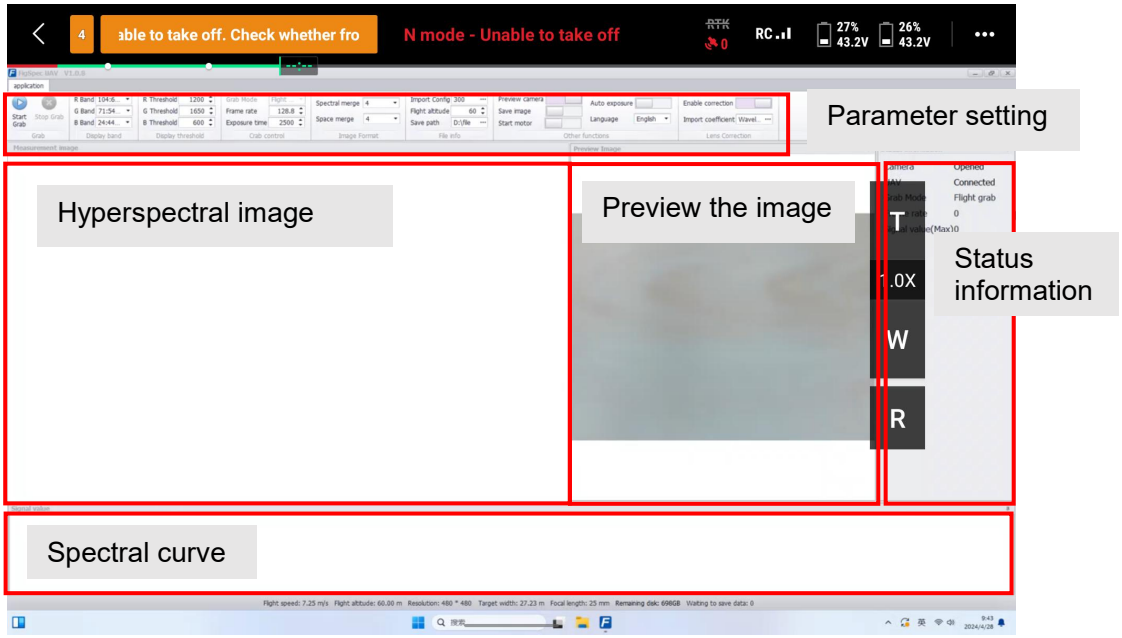


- Easy to operate without the need for professional drone piloting skills, allowing for single-person operation.
- Through the ground station, real-time monitoring of aircraft sampling locations is possible. Additionally, the ground station allows for the preview and correction of waypoint collection data. It supports batch processing for radiance correction, reflectance correction, and regional correction.
- Real-time computation of commonly used vegetation indices.
- Support for custom real-time analysis model input functionality.
- ENVI is perfectly compatible with multiple data formats.

7. Software Introduction

7.1. Introduction to FigSpec UAV Acquisition Software

Software interface and parameter description



The FigSpec UAV interface is divided into six main parts：

1. Parameter Settings Column:

This column contains settings for all parameters used by the camera.

2. Hyperspectral Image Display Area:

Generates false-color hyperspectral images based on the selected RGB channels and RGB thresholds in the parameters.

3. Preview Image Display Area:

Real-time display of the camera's preview image in pseudo-color.

4. Camera Status Bar:

Displays the camera's connection status, acquisition mode, and acquisition status information.

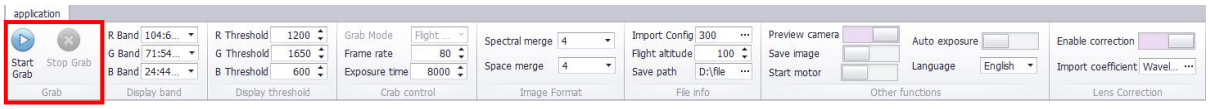
5. Spectral Curve Display Area:

Real-time display of the spectral curve of the point with the highest signal level on the latest data.

6. Flight Parameters Column:

Display of flight parameters, which are the parameters applied during actual task execution (calculated based on the parameters set in the parameter settings column).

7.2. Introduction to Parameter Settings Column

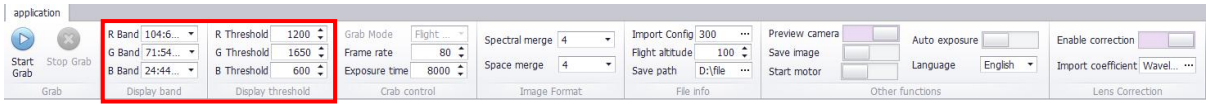


1. Start Crab:

Initiates the collection of hyperspectral images. (Shortcut key set to C1)

2. End Crab:

Stops the collection of hyperspectral images. (Shortcut key set to C2)

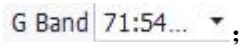


3. RGB Band Settings

For cameras with a wavelength range of 400-1000nm, the recommended setting is:

R:104-610nm 

;

G:71-542nm 

;

B:24-443nm 

;

The pseudo-color image synthesized from these three channels closely resembles true-color imagery.

4. RGB Threshold Settings

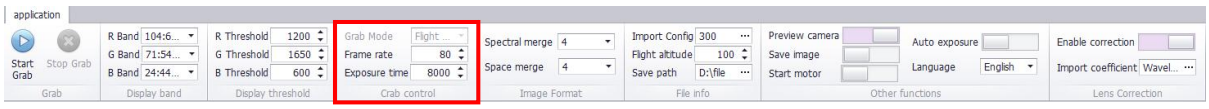
RGB threshold is used to adjust the visual effect of hyperspectral images.

If the image appears too dark, lower the threshold.

If the image appears too bright, raise the threshold.

For cameras with a wavelength range of 400-1000nm, the approximate scale is recommended to be:

R:G:B = 5:6:4； Such as： R:G:B = 800:1000:600



5. Parameter Settings

1) Shooting Mode:

Flight Mode: FS60C defaults to flight mode and cannot be modified (unmanned aerial vehicle push-broom collection mode).

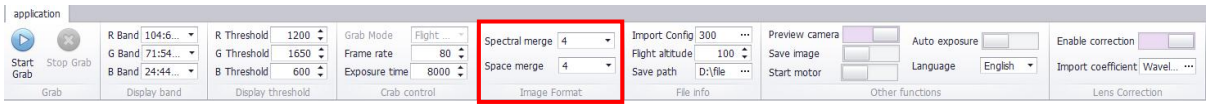
2) Frame Rate:

The speed at which the camera captures images per second, measured in Hz. It indicates how many frames the camera captures per second.

3) Exposure Time:

The duration of time the camera measures light exposure for one frame, measured in microseconds (us).

Exposure time needs to be adjusted based on the actual lighting conditions during data collection. The specific tuning process will be explained later.



6. Image Format Settings

1) Spectral merge:

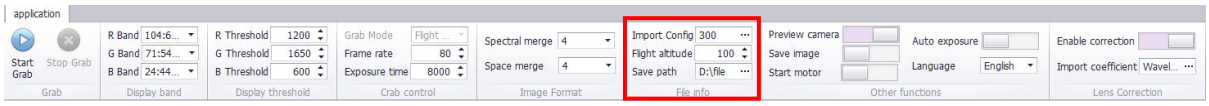
Fusion of spectral channels, i.e., the number of spectral bands of data collected at each pixel. See the detailed parameters in the table below.

2) Space merge:

Fusion of spatial pixels in the image, i.e., the number of pixels in the image's height dimension. See the detailed parameters in the table below.

Here, only the height resolution of the image is listed. The length of the image is derived from the push-broom process.

	Merge mode	1 Merge	2 Merge	4 Merge
FS60C Space Merge	Summation	1920 px	960 px	480 px
FS60C Spectral Merge	Summation	1200	600	300



7. Flight Parameter Settings

1) Import Wavelength:

Each camera has a unique configuration file, and configuration files cannot be shared between cameras. Default configurations are set at the factory and backed up in the camera, with the file format ".chnwave".

2) Flight Altitude:

The altitude at which the drone flies, measured in meters (m).

Flight altitude should be set according to individual needs. It affects flight speed and target resolution (i.e., the actual spatial size corresponding to a single pixel).

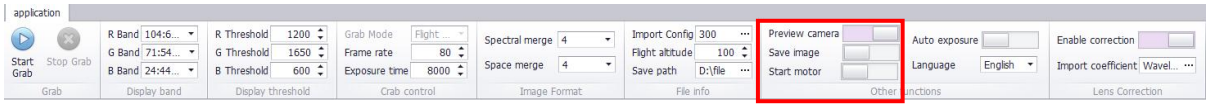
The flight altitude within the drone's mission route needs to be consistent with this parameter.

3) Save Path:

The “Save Path” is the directory where hyperspectral and preview images are saved. Default configurations are set at the factory, typically in the ‘C:\FigSpec\file’ or ‘D:\FigSpec\file’ folder.

Data is saved after completing a full collection cycle (from start to stop).

Note: Data will not be saved if collection is started but not stopped.



8. Other Function Settings

1) Preview Camera:

Default: Enabled. Shows real-time display of the area captured by the camer.

2) Save Images:

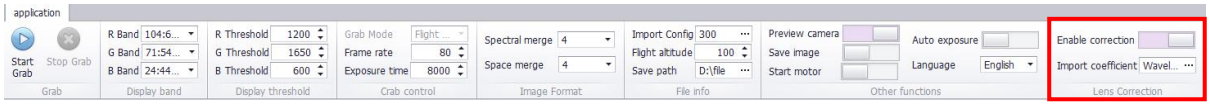
Default: Disabled. Saves images captured by the preview camera. Capture rate is 1 image per second.

3) Enable Motors:

Default: Disabled.

FS60C, FS50 series, and FS62C do not support enabling motors.

Custom onboard cameras (M600PRO mounted on 23 or 27) support enabling motors (i.e., support hover mode).



9. Lens Calibration

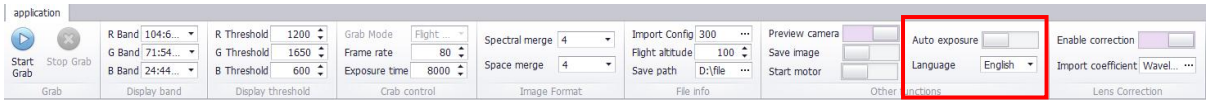
1) Enable Calibration:

Lens calibration corrects the effects of light on the camera itself.

Default: Enabled. Corresponding lens calibration coefficients are imported by default.

2) Import Coefficients:

Import calibration coefficients for the lens. Default configurations are set at the factory and backed up in the camera.



10. Other Parameter Settings

1) Auto Exposure:

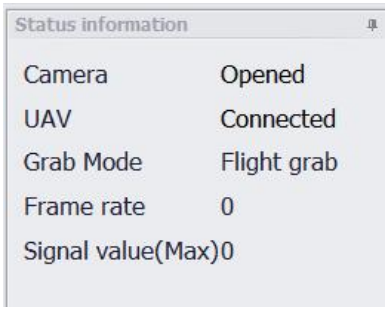
Default: Disabled. To prevent accidental triggering during flight, it is recommended not to enable this feature.

When enabled, clicking on auto exposure on the remote controller will automatically calculate and set the appropriate exposure time for the camera, eliminating the need for manual adjustment.

2) Language selections:

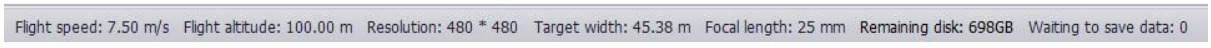
Allows switching between Chinese and English language systems. Default setting is based on the system language.

7.3. Introduction to Status Information



1. Camera: Displays the Connection Status of the Camera
   1) If enabled, it can be used.
   2) No connection indicates that it cannot be used.
2. UAV: The Connection Status of a Drone
   1) Connected means the camera can be controlled from the drone.
   2) Not connected means the camera cannot be controlled from the drone.
3. Grab Mode:
   Displays the currently selected shooting mode.
4. Frame Rate
   Display real-time acquisition frame rate.
5. Signal Value (max) :
   Displays the maximum value of the semaphore in the real-time collected data, in %.

7.4. Flight Parameters



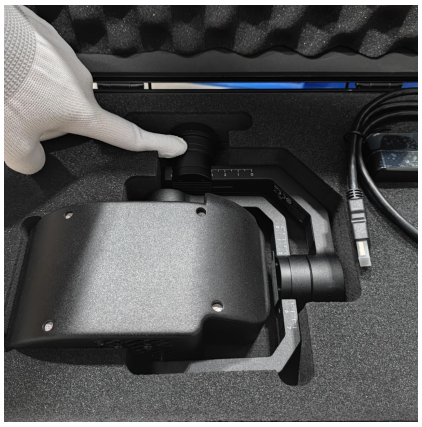
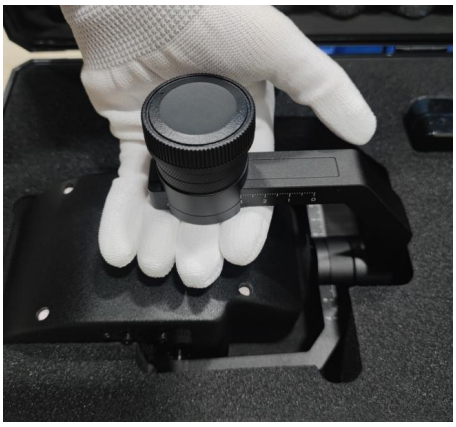
After setting parameters in the parameter setting bar, the software automatically calculates the standard flight parameters and displays them in the bottom bar of the software:

- Flight speed m/s: The recommended flight speed of the drone, the drone planning route tasks, the speed needs to be consistent with the speed here.
- Flight altitude m: The flight altitude within the UAV route task needs to be consistent with the parameters here.
- Resolution px: mainly refers to the height of the hyperspectral image.
- Target width(m): The actual width that the camera can capture.
- Focal length(mm): The default setting is based on the camera configuration and cannot be changed.
- Remaining disk(MB): indicates the disk space. If the disk space is insufficient, the storage fails.
- Waiting to save data: Displays the amount of data that has not been saved. If 0 is displayed, all data is saved.

8. Demonstration of Real Flight Process Operations

8.1. Camera Installation

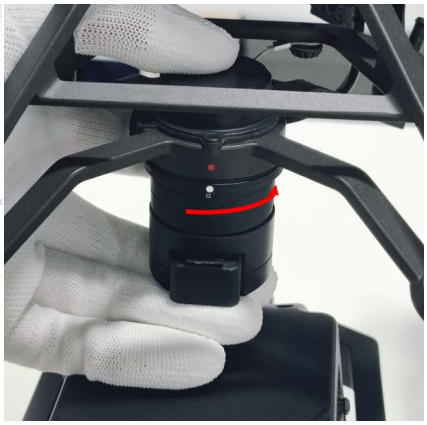
Grab the top of the head, hold the top of the head, take out the camera, as shown in the picture.

Remove lens protection cap.

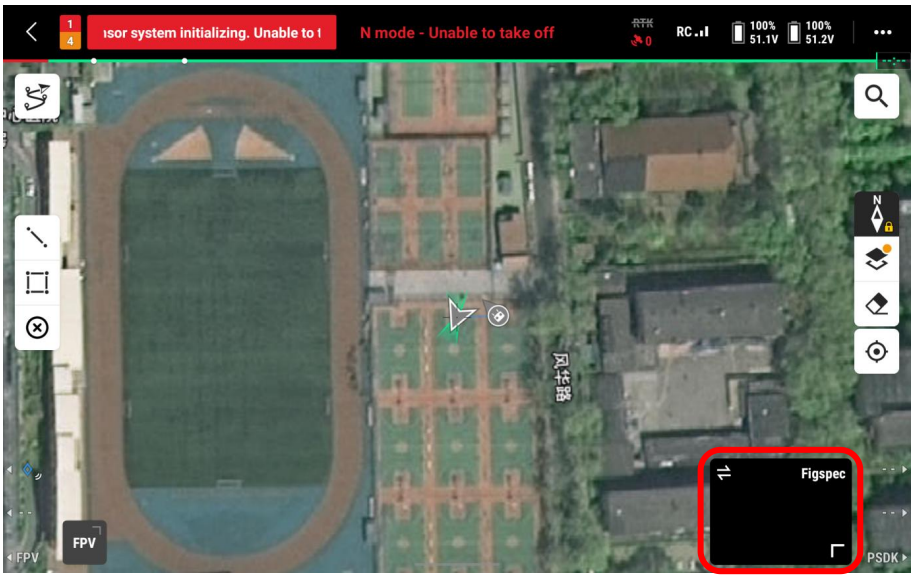


Turn the head protective cover counterclockwise and take it off. Put the white point toward yourself and buckle it onto the UAV head holder. Rotate the head holder so that the red point of the head holder is aligned with the red point of the UAV head holder.

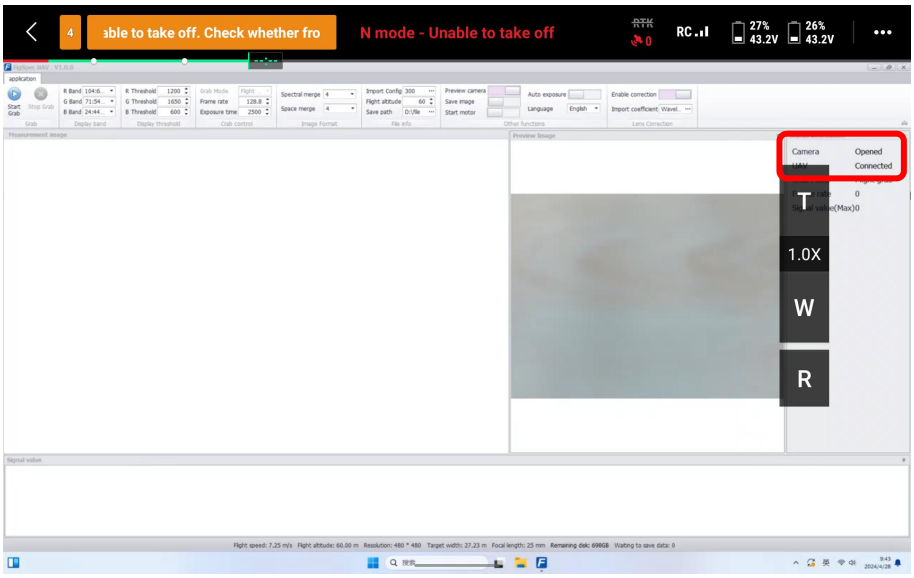



8.2. Camera and Drone Power On

1. Turn on the drone remote control and turn on the drone.
2. Wait for the camera autobiography to complete and return to the middle position, and press the camera switch within 5s.
3. After the camera is turned on, a small window named "Figspec" will appear in the lower right corner of the remote control.

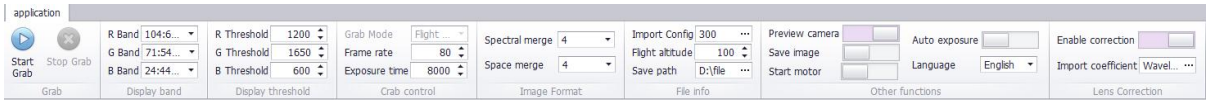


When the camera is started, the software will start automatically. After the software starts, the status bar on the right side shows that the camera and the drone are connected, indicating that the camera has been started correctly.



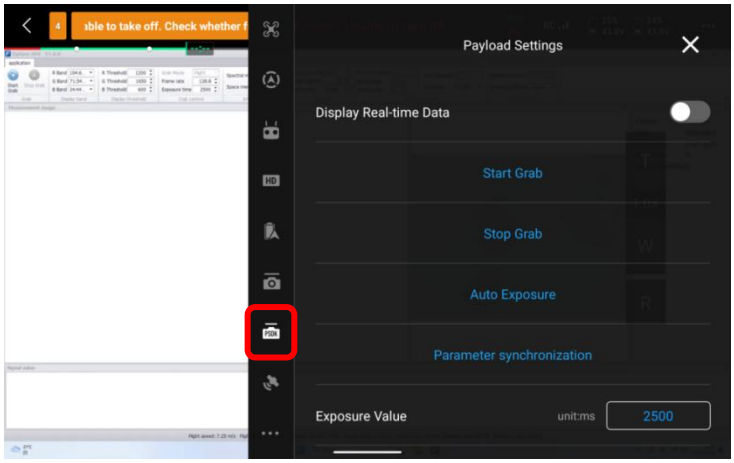
8.3. Data Collection Parameter Settings

The preset parameters are as shown in the figure.



Setting mode:

1. Click the three points in the upper right corner of the remote control to find PSDK, as shown in the figure

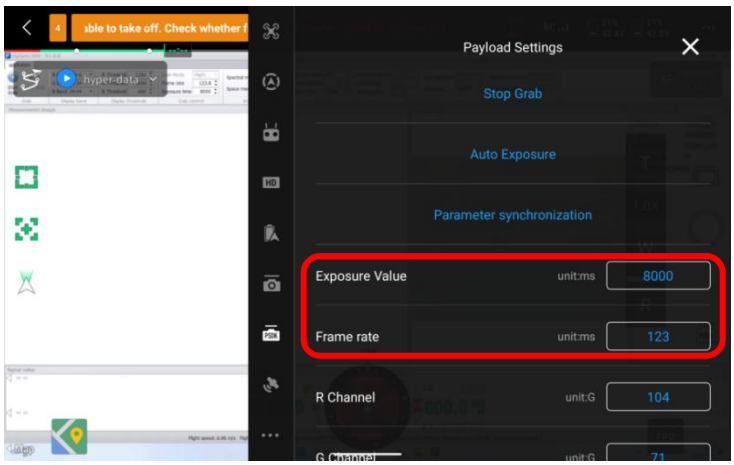


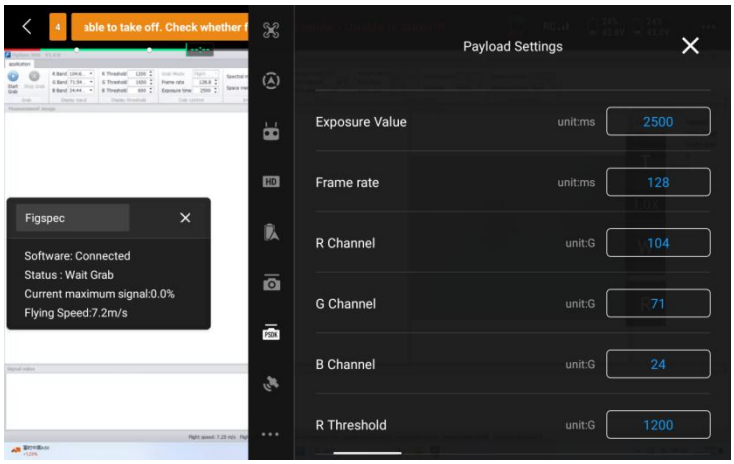
2. Drag up in PSDK and select parameters to set

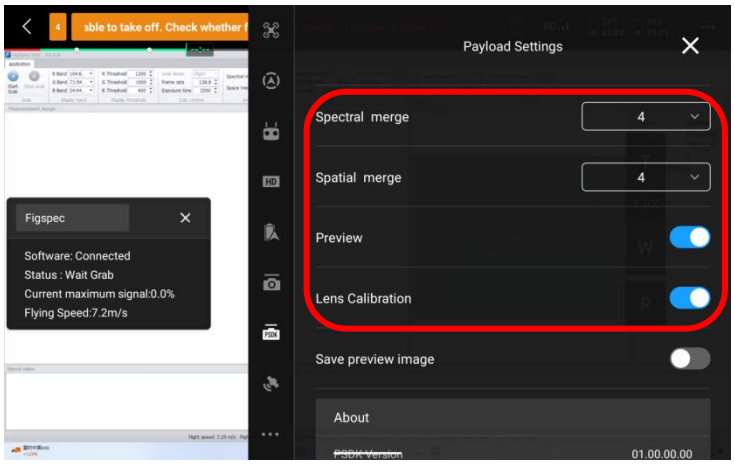
Mainly set the exposure value, frame rate, flight altitude, spectral merge, spatial merge, and confirm that the preview camera and lens calibration options are turned on (these parameters cannot be modified during the acquisition task).

Note: First of all, adjust the exposure value. After the exposure is adjusted, then set other parameters.

The RGB threshold and channel are not set, but can be modified during the collection task.

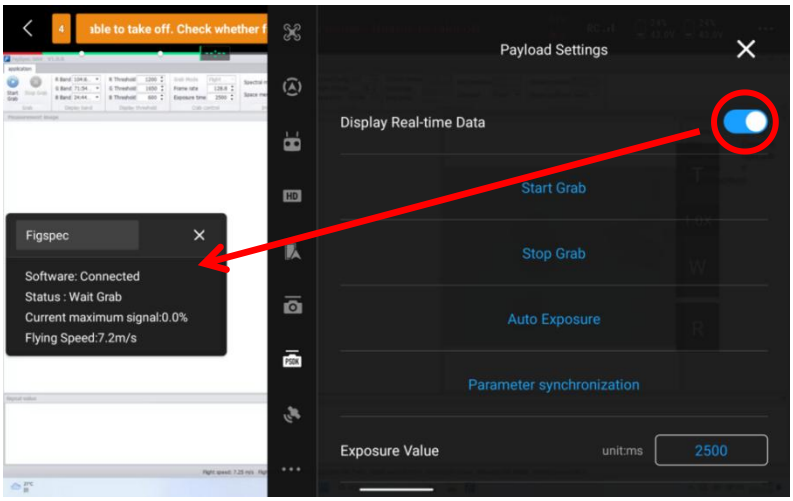






Display Real-time Data

After opening, the Figspec window pops up, showing the software connection status, camera acquisition status, maximum signal (adjusting exposure time), recommended flight speed (planning route).

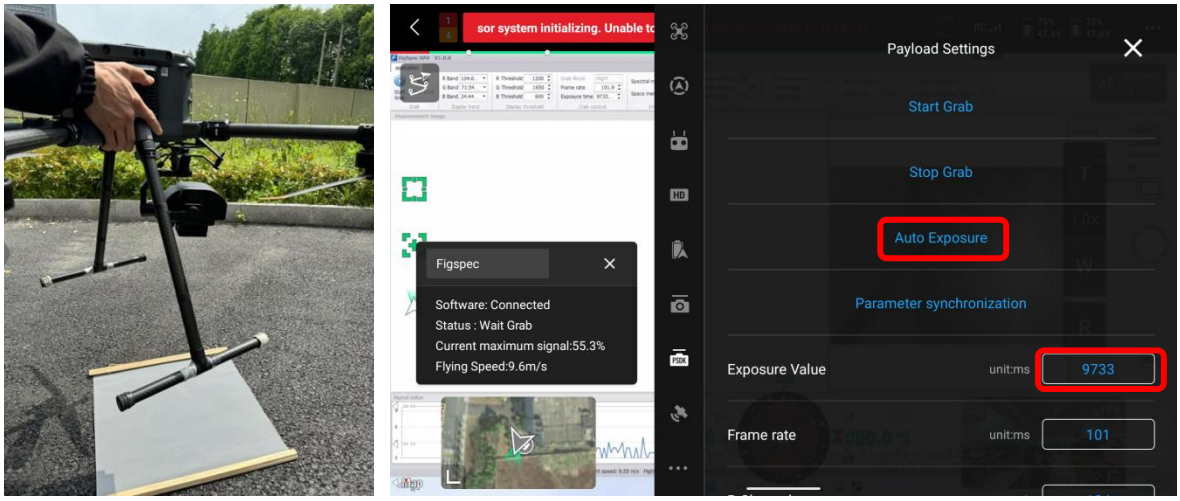


1. First, adjust the Exposure Value:

Auto Exposure adjusts the semaphore of the Reflectivity Calibration Cloth to about 80%.

Place the "reflectivity calibration cloth" on the ground, lift the drone so that the camera is directly above the "reflectivity calibration cloth".

There is no shadow in the lower part of the camera. Click on the “Auto Exposure”, and the software will adjust the exposure value by itself to make “Current maximum signal” reach about 80%.



2. Then set other parameters.

- Frame rate:1-128fps

  In order to avoid the drone flying too fast, it is recommended to set the “Frame rate” between 50-100.

- Flying Height:

  Set it up according to your own needs while ensuring safety.

  The recommended“Flying Height”is 50-150 meters.

- Spectral merge and Spatial merge:

  Set up according to your own needs.

  It is recommended to set to 4 or 2.

Note:

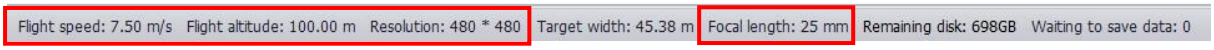
- If the wind speed is too high, please reduce the frame rate or flight altitude.
- If the wind speed is greater than level 4, it is not recommended to carry out collection.

Now set it according to the following parameters:

Frame rate: 80fps Flying Height: 100m

Spectral merge: 4 Spatial merge: 4

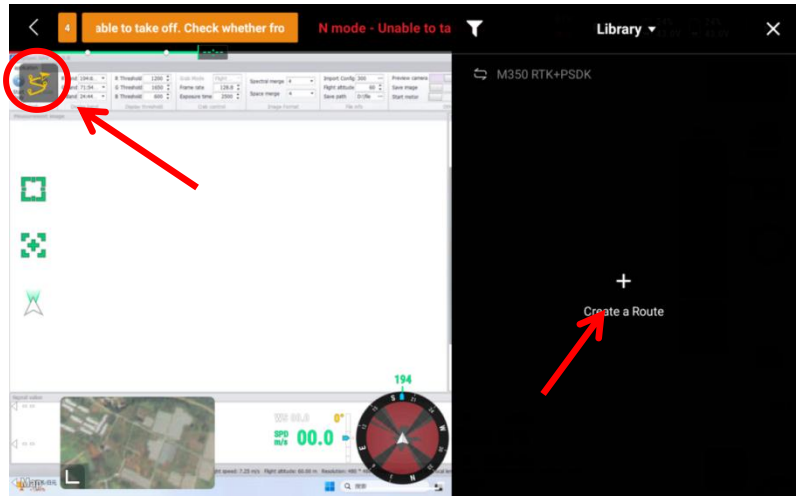
According to the software parameter Settings, the bottom of the software will calculate the appropriate mapping aerial photography parameters



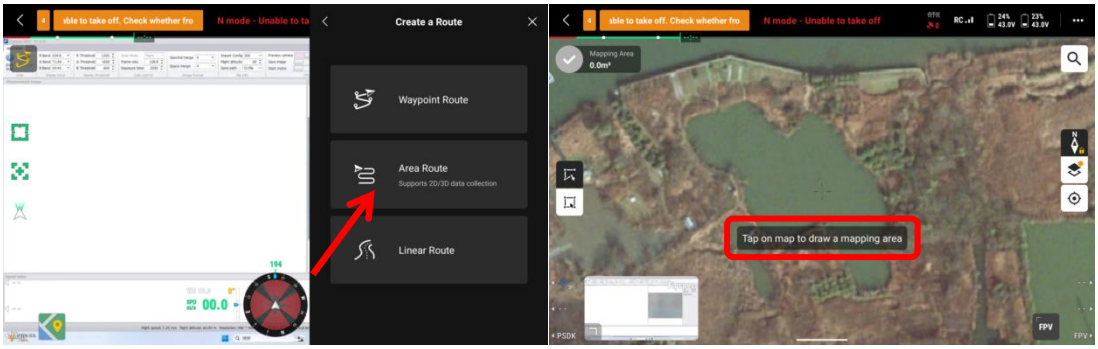
Next, we will plan the route according to the parameters here.

8.4. Plan Flight Route and Set Parameters

Open "Map" on the remote control, click the icon in the upper left corner to create a new route task, and Click "Create a Route" on the right.



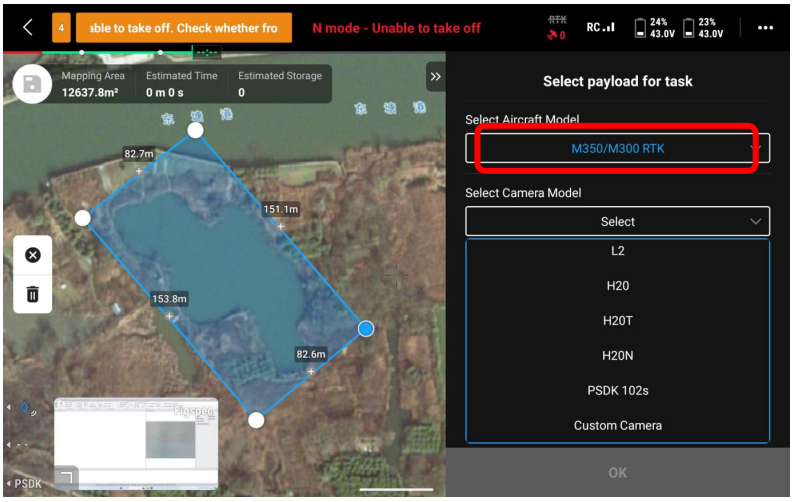
Select“Area Route”.



Click 4 points in the map, and then click "√" in the upper left corner.

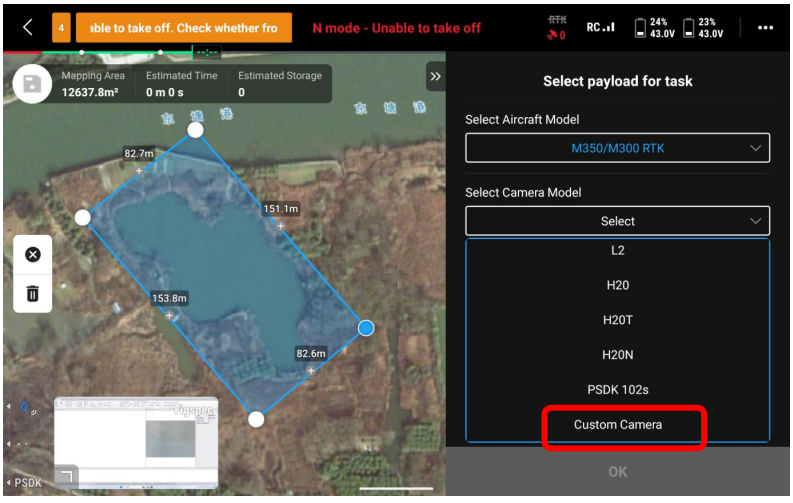


Select M350/M300 RTK.

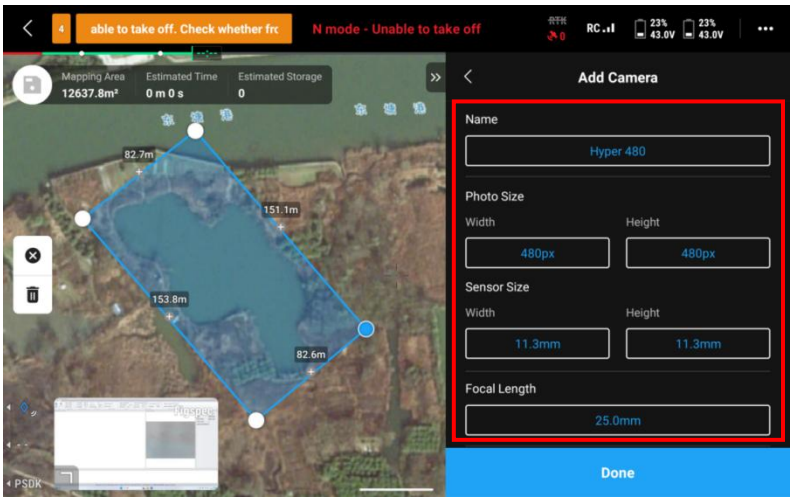


Add a camera:

Click "Custom Camera" in the figure.

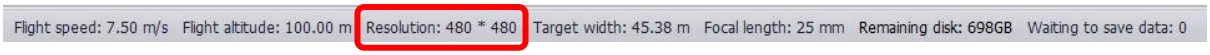


Set camera parameters.



Parameters:

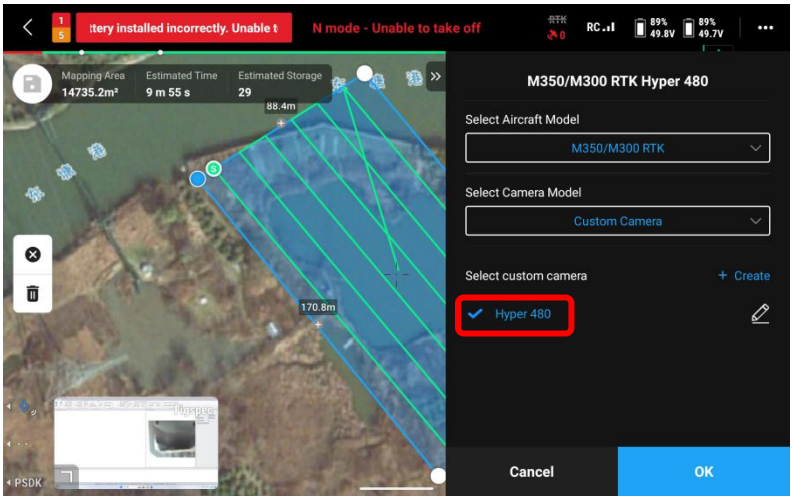
- Name: named according to camera type and resolution.
- Photo Size: The resolution is filled in according to the parameters calculated under the software in the camera.



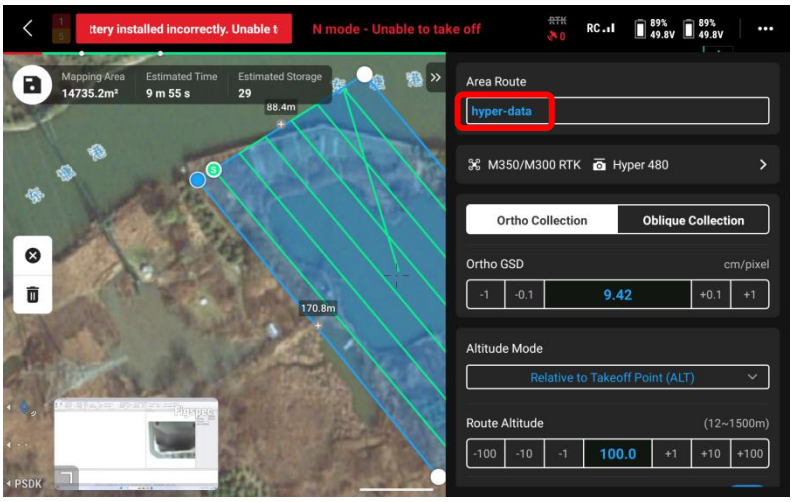
- Sensor Size: The size of the Sensor is fixed 11.3mm.
- Focal Lenth: 25mm.

Click "Done" after the parameter setting is completed.

Select the added camera.Then click "OK".



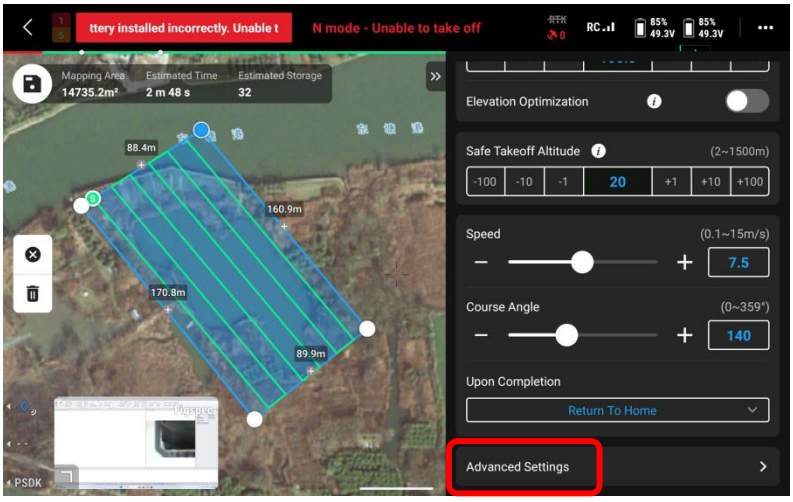
Name the “Area Route”.



Set the “Route Altitude”.And turn off“Elevation Optimization”.



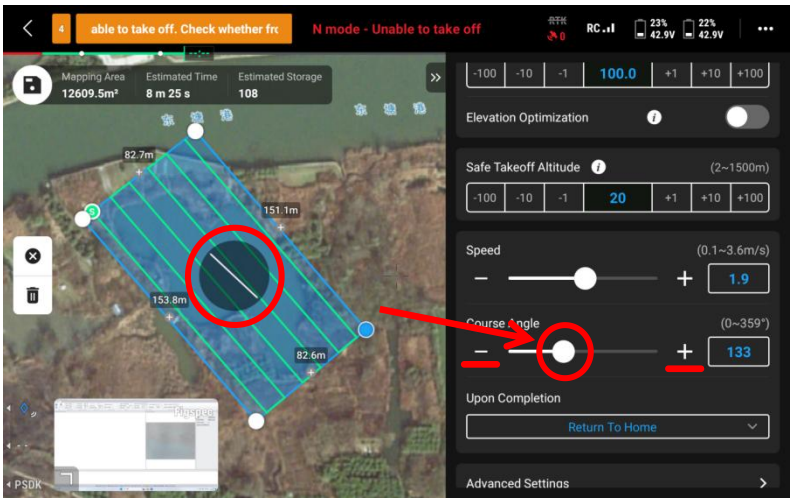
Then click “Advanced Settings” at the bottom.



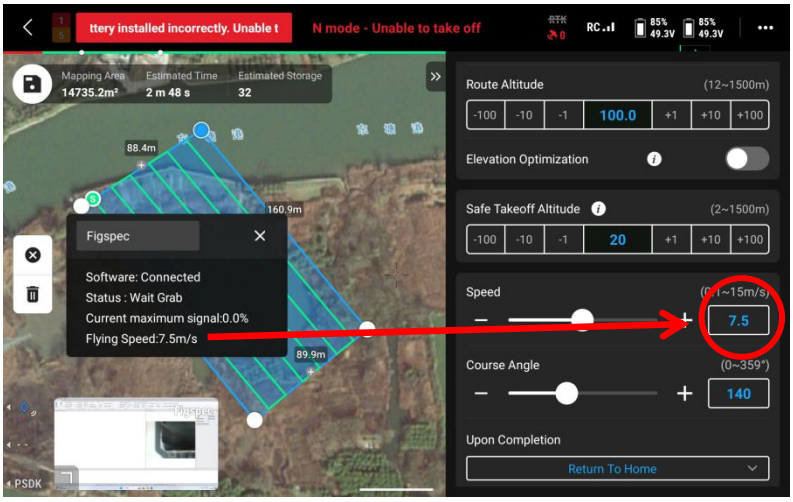
Set the “Frontal Overlap Ratio” to 10%. Then return to the previous interface.



Adjust the route angle so that the direction of the route is the direction of the long side of the blue edge. Fine-tune the angle through "+" and "-".

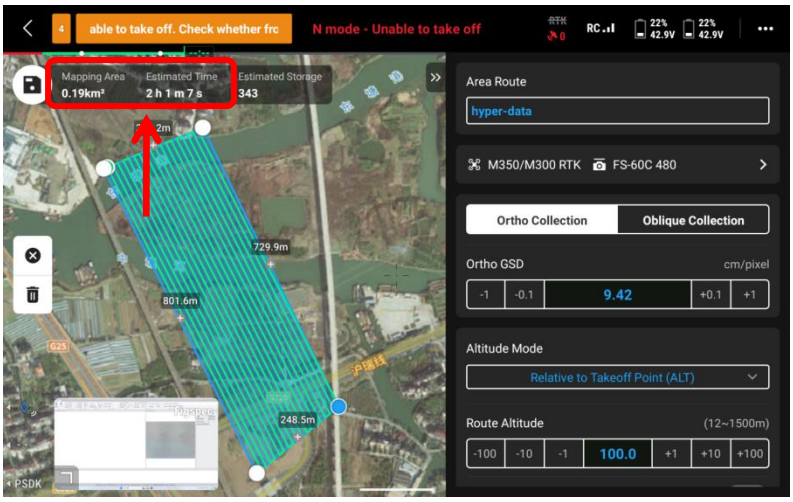


Then set the route speed, which needs to be consistent with the speed recommended by the camera.



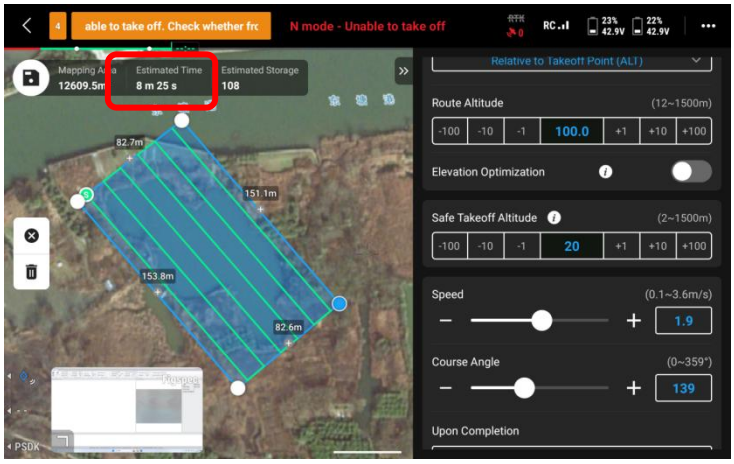
At this point, the route parameters have been set.

According to the route parameters, the interface shows the time required to execute the route and the size of the line-bearing area.



We need to resize the area to adjust the time to perform the task.

Adjust the“Estimated Time”to cary out the task to less than 35 minutes.



After confirming the parameters, save the route.



8.5. Real Flight Operations

A. Check drone status before takeoff

B. Check the drone wing fastener

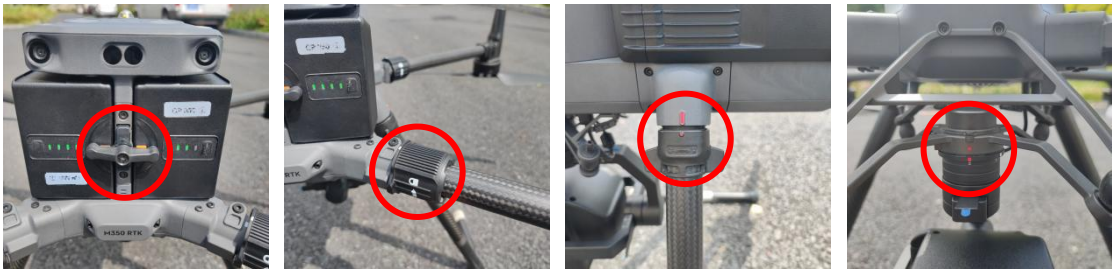
C. Check the drone tripod fastener

D. Check the drone battery retainer

E. Check for obstacles directly above the drone

F. Check for obstacles within 3 meters of the drone

G. Open RTK in remote control

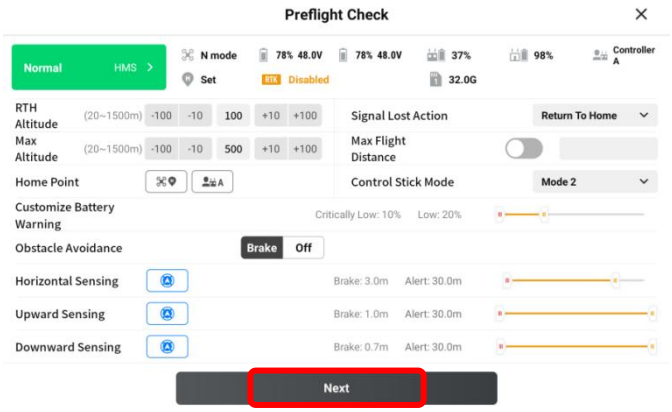


Start the route mission

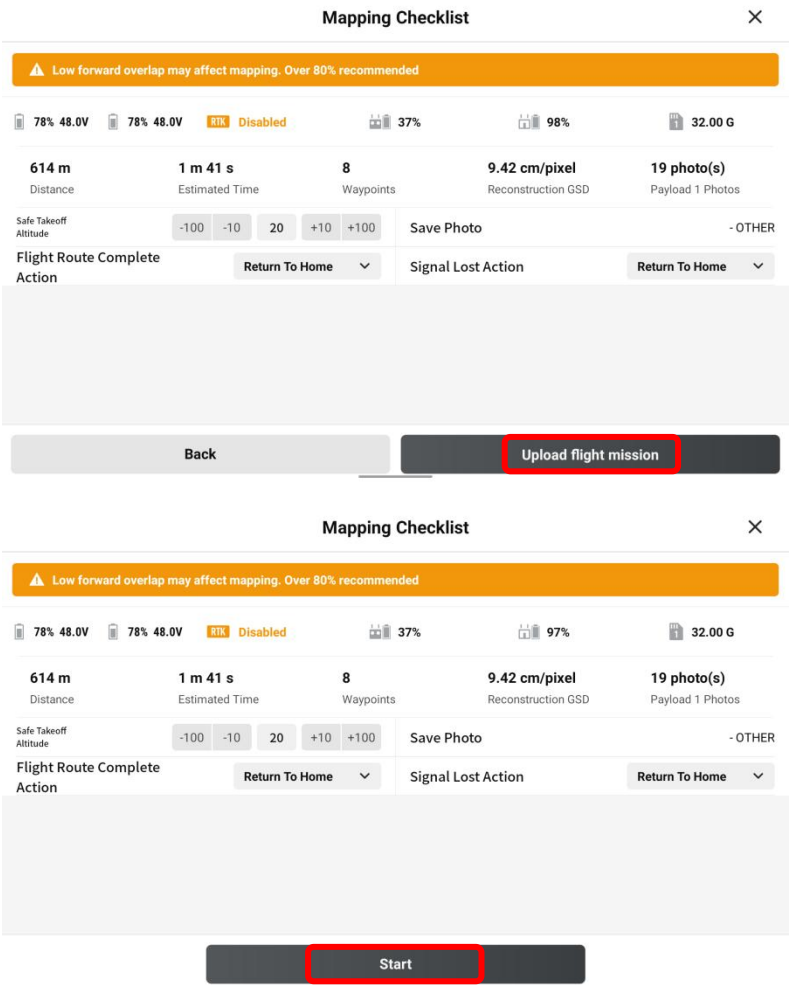
Click the icon in the upper left corner to start the route task.



Check the status of the drone and click "next" if there is no problem.



Then click “Upload flight mission” – “Start”.

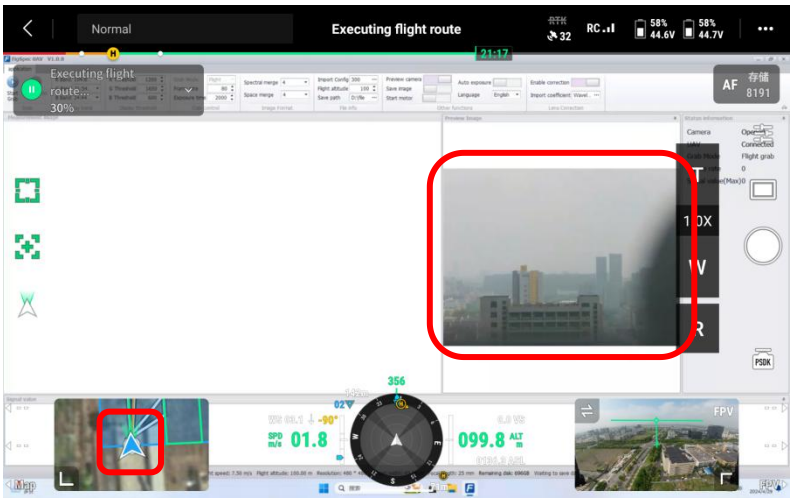


When taking off, the human eye pays attention to the take-off state of the drone, and constantly observes whether the flight path of the drone is safe. When the drone flies far away, the remote control is observed to determine whether the status of the drone is normal, and the interface is switched to the camera interface to observe and preview the camera.





Upon reaching the starting point of the route, there will be a voice prompt saying “Reach the starting point of the route and start the task”. The camera will rotate 90 degrees upwards, allowing observation of the camera capturing images in the air.

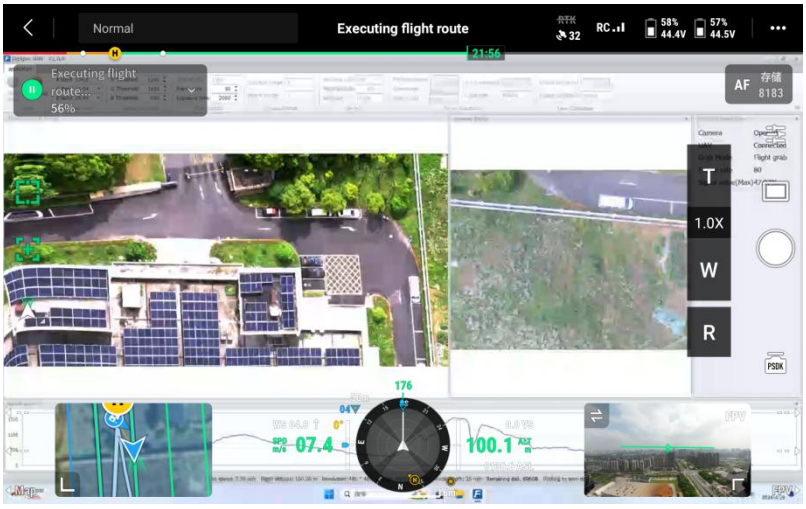


Then click “Start Crab” (C1), and the camera will turn down to carry out the capture task, and the real-time collection of hyperspectral images will be refreshed on the hyperspectral image interface.

When capturing data, the “Reflectivity Calibration Cloth” needs to be placed on the route.



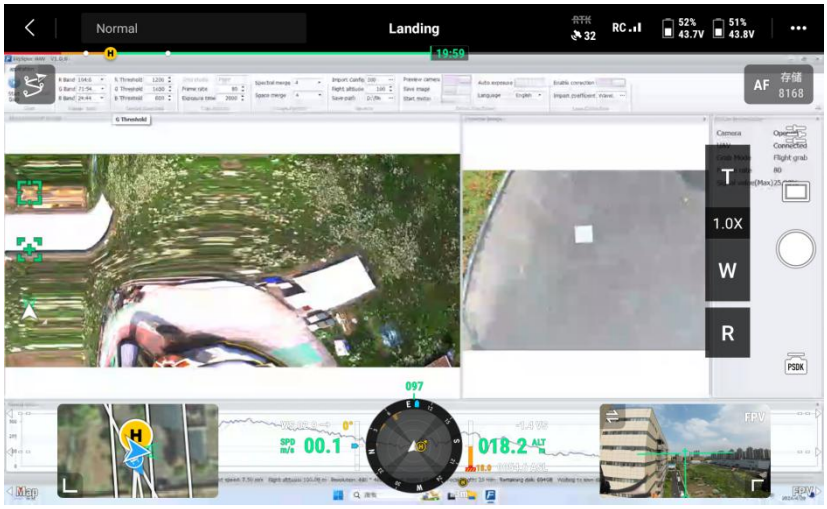
View the picture captured by the camera by observing the remote control interface.



Need to pay attention:

- During the camera collection process, do not accidentally touch "Stop Grab" (C2) and “Automatic Exposure”.(once accidentally touched, you need to re-collect)
- During the acquisition process,the“Exposure Value”, “Frame rate”, “Flying Height”, “Spectral merger”, “Spatial merger”, “Preview”, “LenCalibration” cannot be set.

After the drone returns to the sight of the human eye (within 100 meters), hold the remote control in both hands, and the human eye pays attention to the periphery of the drone to ensure that there are no obstacles, and the drone can land safely.



After the drone returns to the human eye (within 100 meters), hold the remote control with both hands and pay attention to the surroundings of the drone to ensure that there are no obstacles and the drone can land safely;

If you encounter an emergency, click the "H" button in the lower left corner, and then the drone cancels the automatic return and hovers in the air. After the emergency is removed, you can manually control the drone back, or long press the "H" button in the lower left corner, and the drone will continue to return automatically.



When the drone comes back, perform other acquisition tasks or turn off the camera and drone.

Shutdown sequence:

1) Turn off the camera, simply press the power button of the camera, wait for the camera interface to exit completely, then turn off the drone.
2) Turn off the drone.
3) Turn off the remote control.

8.6. Data Copying and Preprocessing

1. Attach an external screen to the camera

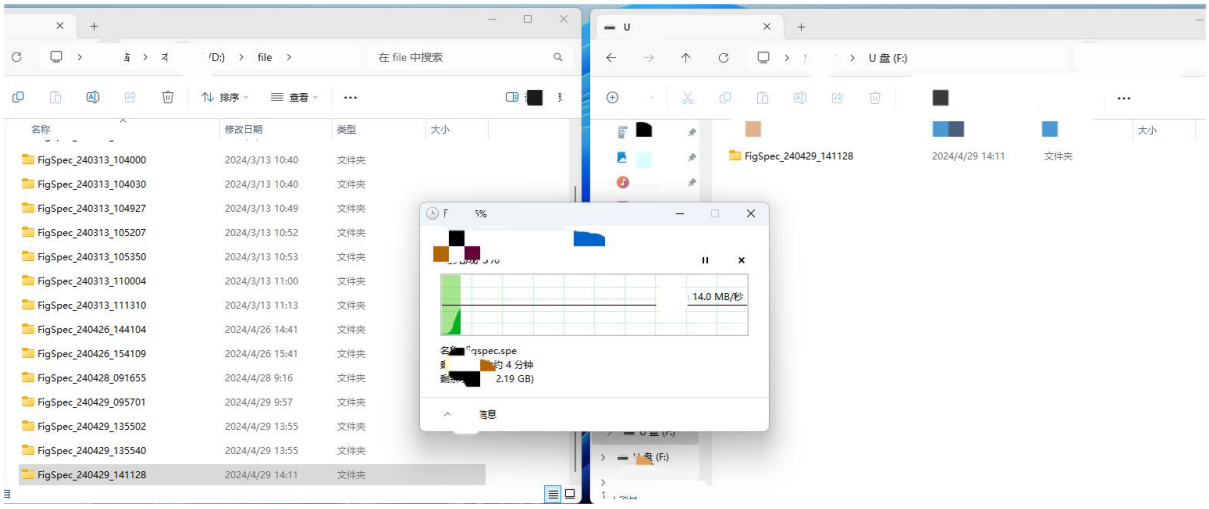


2. External USB splitter, connect the mouse, keyboard, mobile hard disk on the splitter





3. Open the folder where the data is saved and copy the collected data to the mobile hard drive



4. Then the camera is turned off and the data is processed on their own computer. The whole collection of data is over.